Notes on *Statistical Methods for Reliability Data*

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1 Reliability Concepts and Reliability Data

1.1 Introduction

Definition of Reliability: the probablity that a system, vehicle, machine etc will perform its intended function under operating conditions, for a specified period of time.

Reliability is quality over time.

Features of Reliability Data:

1. Typically censored due to the frequent need to analyze life test data before all units are failed.
2. Most reliability data are modeled using distributions for positive random variables such as exponential, Weibull ...
3. Inferences and predictions involving extrapolation are often required.
4. It's often necessary to use past experience or other scientific or engineering judgement to provide infomation as input to the analysis of data or to a decision makeing process.
5. The traditional parameters (such as mean and standard deviation) of a statistical model are typically not of primary interest. Instead, specific measures of product reliability or particular characteristics of a failure time distribution (eg. quantiles, rates) are of more interest.
6. Model fitting requires computer implementation of numerical methods, and often there is no exact theory for statistical inferences, especially with censored data.

1.2 Genral Models for Reliability Data

1.2.1 Target Population or Process

Deming(1975) stated that statistical studies can be broadly divided into two different categories:

1. **Enumerative studies answer questions about *populations* that consist of a finite set of identifiable units**. Typically, the study is conducted by randomly selecting a sample from the

population, carefully evaluating the units in the sample and then making an inference about the larger population from which the sample is taken.

Assumption: the sampling frame accurately represents the units in hte population.

2. **Anlytic studies answer questions about *processes* that generate units or other output over time**. Interest might center on the life distribution of electric motors that will be produced in the future.

   Assumption: The process will behave in the future the same as it has in the past.

1.2.2 Causes of Failure and Degradation Leading to Failure

Many failure modes can be traced to some underlying degradation process. For example, automotive brake pads wear with use.

Traditionaly, most statistical studies of product reliability have been based on failure time data, while the actual level of degradation can also be very helpful, especially when there are few or no failures.

Not all failrues can be traced to degradation, though. Some failures can be caused by sudden accidents.

Understanding the physical and chemical mechanisms and random risk factors leading to failure can suggest methods for eliminating failure modes or reducing the probablity of a failure mode, thereby improving reliability.

1.2.3 Environment Effects on Reliability

Environmental factors play an important role in product reliability. A large proportion of product reliability problems result from unanticipated failure modes caused by environmental effects that were not part of the initial reliability evaluation system.

One challenge of product design is to discover and develop economical means of building in robustness to environmental and other factors that manufactures and users are unable to control.

1.2.4 Time Scale

The life of many products can be measured in more than one scale. For example, life would be measured in number of cycles for factory life tests of products such as wahsing machines and toasters, while time in service data is more commonly available.

The choice of a time for measuring product life is often suggested by an underlying process leading to failure, even if the degradation cannot be observed directly.

There are a number of methods that can be used to handle data with more than one life measurements. For example, we measure battery age and number of charge/discharge cycles. We can 1) estimate the effects of both factors and develop a measure of battery life as a function of both or 2) develop a model that use the number of cycles to help explain the variability in the time to failure.

1.3 Repairable Systems and Nonrepairable Units

There are two situations regarding to failure:

1. The time of failure for nonrepairable units or components, or time to first failure of a system.
2. A sequence of reported system-failure times for a repairable system.

A system may have both repairable and nonrepairable components.

1.4 Strategy for Data Collection, Modeling and Analysis

Reliability studies involving laboratory experimentation or field tracking require carefully planning and excution. Mistakes can be costly in terms of materials and time, and there is also a possibility of drawing erroneous conclusion.

1.4.1 Planning a Reliability Study

1. Define problem to be solved and questions to be answered.
2. Consider the resources available for the study(time, money,equipment etc.).

3. Design the experiment or study with an assessment of precision of estimates as a function of the size of the study. As estimation of precision generally depends on unknow model parameters, assessment will require planning values of unknown population and process characteristics.

   It's often useful to conduct a pilot study to obtain information needed when little is known about the target population of process.

4. In new situations, it's useful, before the test, to conduct a trial analysis of data simulated from a proposed model suggested from available information/previous experience. See more details at Chapter 10 , 20 and section 22.5.

1.4.2 Strategy for Data Analysis and Modeling

General strategy:

1. Begin the analysis by looking at the data without make any distributional or other strong model assumptions, i.e., look at the model without any possible distortion.

   The primary tool for this step is graphical analysis.

2. If possible, fit one or more parametric models to the data for the purpose of description, estimation or prediction.

3. Examine appropriate diagnostics and other use other tools to assess the adequacy of model assumptions before makeing any estimation or prediction.

   Especially when there is little data, it may be difficult to detect departures from model assumptions, but this just means that we have no strong evidence against the assumption but NOT means that the assumption can be trusted.

4. Proceed with caution to esimate parameters and make predictions if step 3 is fine. The estimates and predictions usually come with statistical intervals to reflect uncertainty and variability.

5. Display results of the analysis graphically, including estimates or predictions with uncertainty bounds.

6. It's possible to use the results to draw conclusions about the reliability, contigent on particular model asssumptions.

If assumptions that data provide little information to assess their adequacy, it's useful to vary assumptions and assess the impact of such perturbations have on final answers.
This sensitivity analyses should be reported along with conclusions.

2 Models, Censoring, and Likelihood for Failure-Time Data

2.1 Models for Continuous Failure-Time Process

The most widely used metric for reliability of a product is its failure-time distribution and the most commonly collected reliability data.

Cumulative distribution function, probability density function, survival function and hazard function are used to characterize the probability distribution for failure time T.

1. **Cumulative Density Function(CDF)**: $F(t) = P(T \leq t)$ is the cdf of T. It's the proportion of the units (will) fail before time t.

2. **Probability Density Function(PDF)**: the derivative of cdf. It represents the relative frequency of failure times as a function of time.

3. **Survival Function(SF)**: also called as the reliability function, the complement of the cdf. $S(t) = 1 - F(t)$. It represents the probability of surviving beyond time t.

4. **Hazard Function(HF)**, also called hazard rate, the instantaneous failure rate function.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t < T < t + \Delta t | T > t)}{\Delta t} \tag{2.1}$$

$$= \lim_{\Delta t \to 0} \frac{P(t < T < t + \Delta t)}{\Delta t \cdot P(T > t)} \tag{2.2}$$

$$= \frac{f(t)}{1 - F(t)} \tag{2.3}$$

[4.1] Cumulative Hazard Function: $H(t) = \int_0^t h(x)\mathrm{d}(x)$, and $F(t) = 1 - \exp(-H(t)) = 1 - exp(-\int_0^t h(x)\mathrm{d}(x)$.

[4.2] Average Hazard Rate: $AHR(t_1, t_2) = \frac{\int_{t_1}^{t_2} h(u)\mathrm{d}(u)}{t_2 - t_1} = \frac{H(t_2) - H(t_1)}{t_2 - t_1}$, is the hazard rate over the interval and it's the approximate fraction failing per unit time over the specified interval.

[4.3] Hazard rate in FITs: A fit(failure in time) rate is defined as the hazard function in units of 1/hours, and multiplied by 10^9, it's commonly used in high reliability electronics applications.

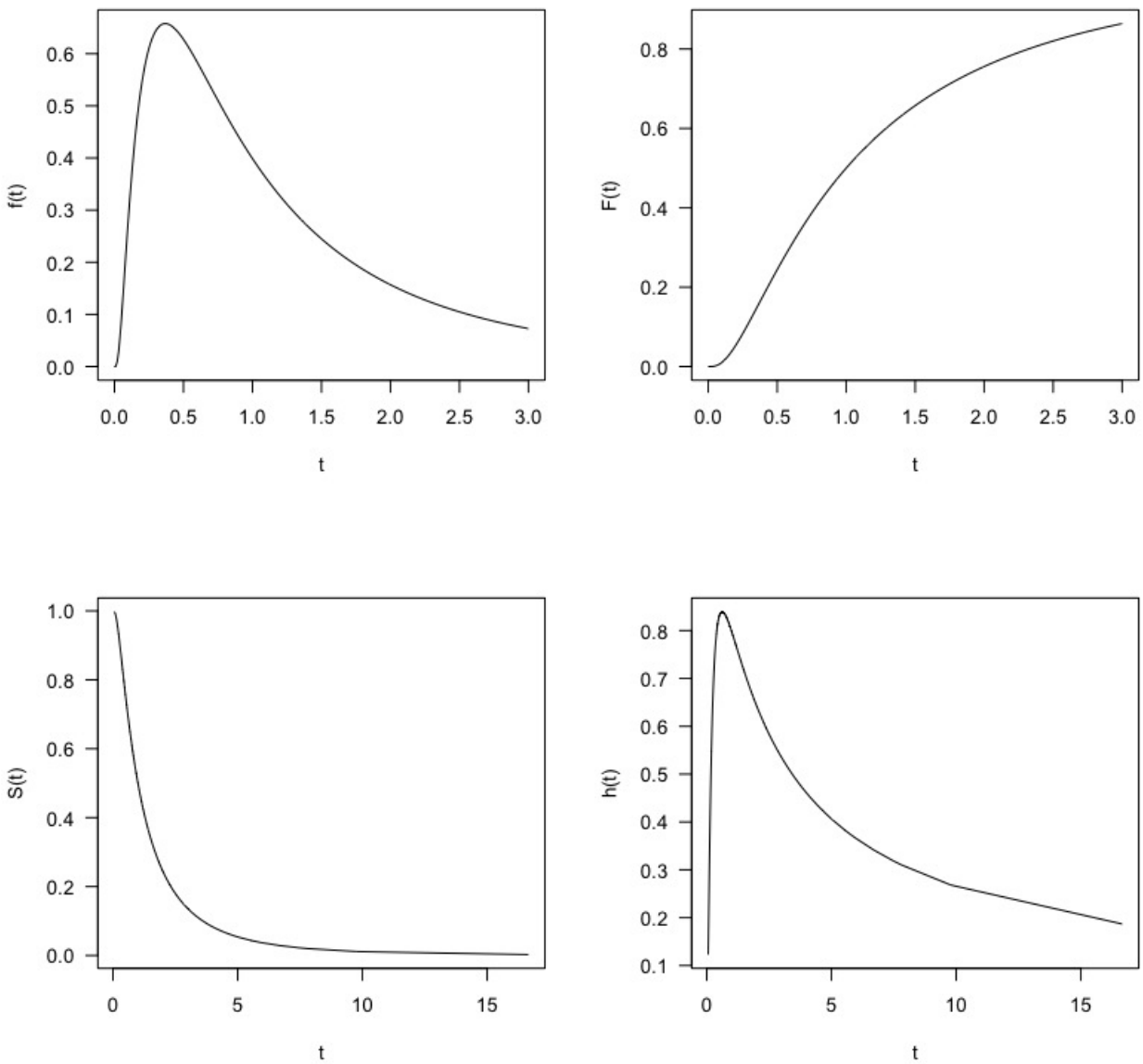

Figure 1: The Four Functions

5. **Quantile Function**:the p quantile of $F(t)$ is the *smallest* t satisfying $P(T \leq t) = F(t) \geq p$.

2.2 Models for Discrete Data From a Continuous Process

1. Partition the time line $(0, \infty)$ into (m+1) **observation** intervals: $(t_0, t_1], (t_1, t_2], \cdots, (t_m, t_{m+1}]$, where $t_{m+1} = \infty and t_0 = 0$.
2. Define multinomial probability that a unit will fail in interval i as $\pi_i = P(t_{i-1} < T \leq t_i) = F(t_i) - F(t_{i-1})$, where $\pi_i \geq 0$ and $\sum_{j=1}^{m+1} = 1$.
3. Then the conditional probability that a unit will fail in interval i given that the unit was still operating at the beginning of the interval i: $p_i = P(t_{i-1} < T \leq t_i | T > t_{i-1}) = \frac{\pi_i}{S(t_{i-1})}$.
4. The survival function is then

$$S(t_i) = \prod_{j=1}^{i} [1 - p_j], i = 1, \cdots, m + 1. \tag{2.4}$$

5. The cdf at time t_i:

$$F(t_i) = 1 - \prod_{j=1}^{i} [1 - p_j] = \sum_{j=1}^{i} \pi_j. \tag{2.5}$$

2.3 Censoring

2.3.1 Censoring Mechanism

Type I censoring:remove all unfailed units from the test at a prespecified time is know as "time censoring" .

Type II censoring: terminate a test after a specified number of failures is known as "failure censoring".

Interval censoring: failures are only observed when inspection conducted, thus interval-censored observations have upper and lower bound on a failure time, also know as inspection data, grouped data, or read-out data.

Random right censoring: some products may have more than one causes of failures and if the primary interest in focused on one particular cause,

then the failures from other causes can , in some situations , be viewed as random right censored data.

Systematic multiple censoring:censoring due to staggered entry of units. The data may be analyzed when not all units have failure, and the data will usually be *multiply right-censored* with some failure times exceeding some of the running times.

2.3.2 Assumptions on Censoring Mechanism

1. The censoring time of a unit depend ONLY on the history of the observed failure-time process.

2. Censoring is non-informative, i.e., the censoring times of units provide no information about the failre-time distribution, this is related to the first assumption.

2.4 Likelihood

The total likelihood , or joint probability of the data for n independent observations including left-censored, interval-censored and right-censored data:

$$L(\mathbf{p}; DATA) = \mathcal{C} \cdot \prod_{i=1}^{m+1} [F(t_i)]^{\ell_i} \cdot [F(t_i) - F(t_{i-1})^{d_i}] \cdot [1 - F(t_i)]^{r_i}. \tag{2.6}$$

where $n = \sum_{j=1}^{m+1}(d_j + \ell_j + r_j)$ and $\mathcal{C}$ is a constant which is usually taken as 1.

Likelihood for Random Censoring in the Intervals

$$\begin{aligned} L_i(\mathbf{p}; data_i) &= \{P[(T \le C) \cap (t_{i-1} < T \le t_i)]\}^{d_i} \{P[(C \le T) \cap (t_{i-1} < C \le t_i)]\}^{r_i} \\ &= \Big[\int_{t_{i-1}}^{t_i} f_T(t)[1 - F_C(t)]\mathrm{d}t\Big]^{d_i} \times \Big[\int_{t_{i-1}}^{t_i} f_C(t)[1 - F_T(t)]\mathrm{d}t\Big]^{r_i}, \end{aligned}$$

for r_i right censored observations, and d_i failures in $(t_{i-1}, t_i]$.

3 Non-parametric Estimation

3.1 Estimation from Singly Censored Interval Data

The non-parametric estimator $\hat{F}(t)$ based on binomial distribution:

$$\hat{F}(t) = \frac{\#offailuresuptotimet_i}{n} = \frac{\sum_{j=1}^{i} d_j}{n}. \tag{3.1}$$

where n is the initial number of units, d_i is the units that failed/died in interval (t_{i-1}, t_i).

3.2 Basic Ideas of Statistical Inference

The estimates of $\hat{F}(t)$ can be interpreted beyond the particular sample units, and used to make inference about the process or larger existing population of units from which the sample units are choosen randomly (Inferential Statistics).

When repeating the sampling and estimation a lot of times, the sampling distribution of of $\hat{F}(t_i)$ can be produced, which can provide insight into the possible deviation between $\hat{F}(t_i)$ and $F(t_i)$.

Confidence intervals are one of the most useful ways of quantifying the uncertainty due to "sampling error" arising from limited sample sizes.

Converage probability is the probability that a confidence interval procedure will result in an interval containing the quantity of interest.

3.3 Confidence Intervals from Complete or Singly Censored Data

3.3.1 Point-wise Binomial-based Confidence Interval for $F(t_i)$

A conservative $100(1-\alpha)\%$ confidence interval $[\underset{\sim}{F}, \tilde{F}]$ is:

$$\underset{\sim}{F}(t_i) = \left\{1 + \frac{(n - n\hat{F} + 1)\mathcal{F}_{1-\alpha/2;2n-2n\hat{F}+2,2n\hat{F}}}{n\hat{F}}\right\}^{-1}. \tag{3.2}$$

$$\tilde{F}(t_i) = \left\{1 + \frac{(n - n\hat{F})}{(n\hat{F} + 1)\mathcal{F}_{1-\alpha/2;2n\hat{F}+2,2n-2n\hat{F}}}\right\}^{-1}. \tag{3.3}$$

where $\hat{F}$ is $\hat{F}(t_i)$, $\mathcal{F}$ is the F distribution.

3.3.2 Pointwise Normal-Approximation Confidence Interval for $F(t_i)$

An approximate $100(1-\alpha)\%$ confidence interval for $F(t_i)$ is

$$[\underset{\sim}{F}(t_i), \tilde{F}(t_i)] = \hat{F}(t_i) \pm z_{1-\alpha/2}\hat{se}_{\hat{F}}. \tag{3.4}$$

where $\hat{se}_{\hat{F}} = \sqrt{\hat{F}(t_i)[1-\hat{F}(t_i)]/n}$.

3.4 Estimation from Multiply Censored Data

The size of the risk set at the beginning of interval i is:

$$n_i = n - \sum_{j=0}^{i-1} d_j - \sum_{j=0}^{i-1} r_j \tag{3.5}$$

where d_j is the number of units that died/failed in the jth interval $(t_{i-1}, t_i]$, and r_j is the number of units censored at t_j.

Then the estimator of the conditional probability of failing in interval i given the unit enters this interval:

$$\hat{p}_i = \frac{d_i}{n_i} \tag{3.6}$$

The survival function and non-parametric estimator of $F(t_i)$ can be obtained through (2.4).

3.5 Pointwise Confidence Intervals from Multiply Censored Data

3.5.1 Approximate Variance of $\hat{F}(t_i)$

$Var(\hat{F}(t_i)) = Var(\hat{S}(t_i))$ as they sum to a constant 1. The approximate variance can be obtained by first-order Taylor series expansion:

$$\hat{S}(t_i) \approx S(t_i) + \sum_{j=1}^{i} \left.\frac{\partial S}{\partial q_j}\right|_{q_j} (\hat{q}_j - q_j), \tag{3.7}$$

where $q_j = 1 - p_j$ and q_j values are approximately uncorrelated binomial **proportions**.

Then

$$Var(\hat{F}(t_i)) = Var(\hat{S}(t_i)) \approx [S(t_i)]^2 \sum_{j=1}^{i} \frac{p_j}{n_j(1-p_j)}. \tag{3.8}$$

Note the detailed derivation of 3.8 is, for a single j^{th} term of the summation, we have $\frac{\partial S}{\partial q_j} = \prod_{j=1}^{i-1} q_j$ and the variance for this specific term is $(\frac{\partial S}{\partial q_j})^2 Var(\hat{q_j}) = (S(t_{i-1}))^2 \frac{q(1-q)}{n} = (S(t_{i-1}))^2 \frac{q^2(1-q)}{nq} = (S(t_i))^2 \frac{1-q}{nq}$.

3.5.2 Greenwood's Formula

Substitute p_j, $S(t_i)$ by $\hat{p}_j$, $\hat{S}(t_i)$ respectively in (3.8) gives the Greenwood's formula.

3.5.3 Pointwise Normal-Approximation Confidence Interval for $\hat{F}(t_i)$

A normal approximate $100(1-\alpha)\%$ confidence interval

$$[\underset{\sim}{F}, \tilde{F}] = \hat{F}(t_i) \pm z_{(1-\alpha/2)} \hat{se}_{\hat{F}}, \tag{3.9}$$

and it's based on the assumption that $Z_{\hat{F}} = \frac{\hat{F}(t_i) - F(t_i)}{\hat{se}_{\hat{F}}}$ can be approximated by a standard normal distribution.

When the sample size is not large and the data may be heavily skewed, a logit transformation may be helpful for a better approximation as $log(\hat{F}(t_i))$ may behave more like a standard normal random variable, i.e.,

$$Z_{logit(\hat{F})} = \frac{logit(\hat{F}(t_i)) - logit(F(t_i))}{\hat{se}_{logit(\hat{F})}}. \tag{3.10}$$

3.6 Estimation from Multiply Censored Data with Exact Failures

Exact failures arise from a continuous inspection process, while the width of intervals approaches zero, the step functions of $\hat{F}(t)$ increases at the reported failure times. This limiting case of interval-based non parametric estimatro is know as the product-limit or **Kaplan-Meier estimator**.

3.7 Simulataneous Confidence Bands

The motivation is to quantify the sampling uncertainty simutaneously over a range of time t instead of pointwise confidence intervals at particular specified value of t.

Approximate $100(1-\alpha)\%$ simultaneous confidence bands for $F(t)$ can be obtained from

$$[\underset{\sim}{F}(t), \widetilde{F}(t)] = \hat{F}(t) \pm e_{(a,b,1-\alpha/2)} \hat{se}_{\hat{F}(t)}, \tag{3.11}$$

where $t \in [t_L(a), t_U(b)]$ and the approximate factor $e_{(a,b,1-\alpha/2)}$ was computed from a large-sample approximation given in Nair(1984).

3.8 Uncertain Censoring Times

1. If censoring times are random and the form of the distribution is know, a likelihood estimation method could be applied.
2. If censoring pattern and distribution unknown, $\hat{p}_i = d_i(n_i - r_i/2)$ is applied.

4 Loaction-Scale-Based Parametric Distributions

4.1 Quantities of Interest in Reliability Applications

1. *Probability of failure* $p = P(T \leq t) = F(t; \boldsymbol{\theta})$ at a specified t.
2. p *quantile* of the distribution of T with $t_p = F^{-1}(p; \boldsymbol{\theta})$.
3. *hazard function* with $h(t) = \frac{f(t;\boldsymbol{\theta})}{1-F(t;\boldsymbol{\theta})}$.
4. *average/expectation/first moment*

$$\begin{aligned} E(T) &= \int_0^\infty t f(t; \boldsymbol{\theta}) \mathrm{d}t && (4.1) \\ &= -\int_0^\infty t \mathrm{d}(1 - F(t; \boldsymbol{\theta})) && (4.2) \\ &= -tF(t; \boldsymbol{\theta})\big|_0^\infty + \int_0^\infty (1 - F(t; \boldsymbol{\theta})) \mathrm{d}t && (4.3) \\ &= \int_0^\infty (1 - F(t; \boldsymbol{\theta})) \mathrm{d}t && (4.4) \end{aligned}$$

is a measure of the center of $f(t;\boldsymbol{\theta})$.

One possible confusion is if we rewrite $f(t;\boldsymbol{\theta})$ in terms of the hazard function above, the equation will be $E(T) = \int_0^\infty [1 - F(t)]th(t)\mathrm{d}t$. However, these equations are equivalent: because $1 - F(t) = e^{-H(t)}$, $H(t) = \int_0^t h(x)\mathrm{d}x$ and $\mathrm{d}(e^{-H(t)}) = -e^{-H(t)}\mathrm{d}(H(t)) = -e^{-H(t)}h(t)\mathrm{d}t$,

$$
\begin{aligned}
E(T) &= \int_0^\infty [1 - F(t)]th(t)\mathrm{d}t && (4.5)\\
&= \int_0^\infty e^{-H(t)}h(t)t\mathrm{d}t && (4.6)\\
&= -\int_0^\infty t \quad \mathrm{d}(e^{-H(t)}) && (4.7)\\
&= -te^{-H(t)}\Big|_0^\infty + \int_0^\infty e^{-H(t)}\mathrm{d}t && (4.8)\\
&= \int_0^\infty (1 - F(t)\mathrm{d}t && (4.9)
\end{aligned}
$$

It's the same.

5. *variance of* T with $Var(T) = \int_0^\infty [t - ET]^2 f(t;\boldsymbol{\theta})\mathrm{d}t$ and standard deviation $sd(T) = \sqrt{Var(T)}$, is the spread of the distribution of T.

6. *coefficient of variation of* T $\gamma_2 = sd(T)/E(T)$, is useful for comparing the relative amount of variability in different distributions. $1\gamma_2$ is the "signal to noise ratio".

7. *standardized third central moment/ coefficient of skewness* with $\gamma_3 = \frac{\int_0^\infty [t-E(T)]^3 f(t;\boldsymbol{\theta})\mathrm{d}t}{[Var(T)]^{\frac{3}{2}}}$.

4.2 Location-scale and Log-location-scale Distributions

Location-scale family distributions can be written as the form:

$$P(Y \leq y) = \Phi(\frac{y - \mu}{\sigma}) \tag{4.10}$$

While for Log-location-scale family random variable T , log(T) belongs to the location-scale family.

4.3 Exponential Distribution

$$F(t;\theta,\gamma) = 1 - e^{-\frac{t-\gamma}{\theta}}. \tag{4.11}$$

Hazard function is constant.
Useful for some kinds of electronic components(for example, capacitors, or high qualify integrated circuits).

4.4 Normal Distribution

$$F(y;\mu,\sigma) = \Phi_{nor}(\frac{y-\mu}{\sigma}). \tag{4.12}$$

Increasing hazard function.

4.5 Log-normal Distribution

$$F(y;\mu,\sigma) = \Phi_{nor}(\frac{log(y)-\mu}{\sigma}). \tag{4.13}$$

Appropriate for time to failure caused by a degradation process with combinations of random rate constants that combine multiplicatively.

Widely used to describe the time to fracture from fatigue crack growth in metals.

4.6 Smallest Extreme Value Distribution

$$F(y;\mu,\sigma) = \Phi_{sev}(\frac{y-\mu}{\sigma}). \tag{4.14}$$

where $\Phi_{sev} = 1 - e^{-e^z}$ in standardized case.
Exponentially increasing hazard function suggests that SEV is suitable for modeling the life of a product that experiences very rapid wearout after a certain age.

4.7 Weibull Distribution

$$F(T \leq t;\eta,\beta) = 1 - e^{-(\frac{t}{\eta})^\beta}, t > 0 \tag{4.15}$$

Relationship with SEV: If T has a Weibull distribution, then
$Y = log(T) \sim SEV(\mu,\sigma)$ with $\mu = log(\eta), \sigma = 1/\beta$.

4.8 Largest Extreme Value Distribution

$$F(y;\mu,\sigma) = \Phi_{lev}(\frac{y-\mu}{\sigma}). \tag{4.16}$$

where $\Phi_{lev} = e^{-e^{-z}}$ in standardized case.
Less commonly used due to the positive probability of negative observations. It can be used as a model for life if σ is small relative to $\mu > 0$.

4.9 Logistic Distribution

$$F(y;\mu,\sigma) = \Phi_{logis}(\frac{y-\mu}{\sigma}). \tag{4.17}$$

where $\Phi_{logis} = \frac{e^z}{1+e^z}$ in standardized form.

The shape is similar to that of the normal distribution except longer tails, however, the *behavior of the hazard function* is different at upper tail. Preferred to the normal distribution because its cdf can be written in a simple closed form, if computation is a concern.

4.10 Log-logistic Distribution

$$F(y;\mu,\sigma) = \Phi_{logis}(\frac{log(y)-\mu}{\sigma}). \tag{4.18}$$

4.11 Generating Pseudo-random Observations from a Specified Distribution

4.11.1 Pseudo-random Observations from Continuous Distributions

Generate $U_1, \ldots, U_n$ random variables from a $UNIF(0,1)$, then $t_{U_i} = F_t^{-1}(U_i)$ is a pseudorandom sample from F_T.

4.11.2 Pseudo-random Observation from Discrete Distributions

Same general idea as Section 4.11.1, but it could be more complicated as sometimes the discrete quantiles cannot be computed directly.

4.11.3 Efficient Generation of Censored Pseudorandom Samples

General Approach based on order statistics
$U_{(i)}$ denotes the ith order statistic from a random sample of size n from $UNIF(0,1)$, then

$$P(U_{(i)} \leq u | U_{(i-1)} = u_{(i-1)}) = 1 - \left[\frac{1-u}{1-u_{(i-1)}}\right]^{(n-i+1)}, u \geq u_{(i-1)} \tag{4.19}$$

By applying the method in Section 4.11.1 , generate a random variable U to replace the P to solve the u as $U_{(i)}$.
Application:

Failure-Censored Samples: generate pseudorandom sample with n units and r failures.

1. Generate $U_1, \ldots, U_{(r)}$(the probability) pseudorandom observations from $UNIF(0,1)$.
2. Find the order statistics $U_{(1)}, \ldots, U_{(r)}$ given $U_{(0)} = 0$. by (4.19).

$$\begin{aligned} U_{(1)} &= 1 - [1 - U_{(0)}] \times (1 - U_1)^{1/n} \\ U_{(2)} &= 1 - [1 - U_{(1)}] \times (1 - U_1)^{1/(n-1)} \\ &\vdots \\ U_{(r)} &= 1 - [1 - U_{(r-1)}] \times (1 - U_1)^{1/(n-r+1)} \end{aligned}$$

3. The pseudorandom sample from $F(t; \boldsymbol{\theta})$ is $T_{(i)} = F^{-1}[U_{(i)}; \boldsymbol{\theta}]$.

Time Censored Sample:
Set t_c is the cutoff time.

1. Generate a new pseudorandom observation U_i from $UNIF(0,1)$. Compute $U_{(i)} = 1 - [1 - U_{(i-1)}] \times (1 - U_i)^{1/(n-i+1)}$ and $T_{(i)} = F^{-1}[U_{(i)}; \boldsymbol{\theta}]$.
2. If $T_i > t_c$ stop; else, $i = i + 1$ and return to step 1.

5 Other Parametric Distributions

The coefficient of variation (CV) is defined as the ratio of the standard deviation σ to the mean μ: $c_{\mathrm{v}} = \frac{\sigma}{\mu}$. It shows the extent of variability in relation to the mean of the population[1].

[1] `https://en.wikipedia.org/wiki/Coefficient_of_variation`

6 Probability Plotting

Probability plots use special scales on which a cdf of a particular distribution plots as a straight line.

6.1 Purpose

1. Assess the adequacy of a particular distributional model.
2. Provide non-parametric graphical estimates of probabilities and distribution quantiles.
3. Obtain graphical estimation of parametric model parameters by fitting a straight line through the points on a probability plot.
4. Display the results of a parametric maximum likelihood fit along with the data.

6.2 Linearing a CDF

By find the transformation of $F(t)$ and t such that the relationship between the transformed variables is linear, one can obtain a linearized plot of time versus the CDF.

6.3 Probability Plotting Positions

6.3.1 Criteria for Choosing Plotting Positions

1. Checking distributional assumptions: The choice of plotting positions, in moderate-to-large samples, is not so important.
2. Estimation of parameters: The "best" plotting positions will depend on the assumed underlying model and the functions to be estimated. For complete data, let i index the ordered observations, then $p_i = \frac{i-0.5}{n}$ is a good choice for general purpose use in probability plotting.
3. Display of maximum likelihood fits with data: The criteria is that the line "fit" the points well when the assumed model being fit with ML agrees with the data.

6.3.2 Choice of Plotting Positions

1. *Continuous inspection data and single censoring*:Let $t_{(i)}$ be the ith ordered failure time, $\hat{F}(t_i) = i/n$ be a step function increasing by $1/n$ at each reported failure time, a reasonable plotting position is the midpoint of the jump:

$$\frac{i-0.5}{n} = \frac{1}{2}\Big[\hat{F}(t_{(i)}) + \hat{F}(t_{(i-1)})\Big] = \frac{1}{2}\Big[\frac{i}{n} + \frac{i-1}{n}\Big]. \tag{6.1}$$

2. *Continuous inspection data and multiple censoring*: due to the multiple censoring, the step increases may be different from $1/n$,

$$p_i = \frac{1}{2}\Big[\hat{F}(t_{(i)}) + \hat{F}(t_{(i-1)})\Big]. \tag{6.2}$$

3. *Interval censored inspection data*: Failures are recorded at the upper endpoint of each interval,

$$p_i = \hat{F}(t_i). \tag{6.3}$$

4. *Arbitrary censored data*: With mixtures of left, right censoring, it requires a compromise between the other two cases.

6.4 Probability Plots with Specified Shape Parameter

There are some distributions not in the location-scale families and cannot be transformed into such as distribution. For example, the gamma and generalized gamma distribution. They may have one or more unknown shape parameters. Two approaches to specify an unknown shape parameter for a probability plot:

1. Plot the data with different given values of shape parameter to find a value that will give probability plot that is nearly linear.

2. Use parametric maximum likelihood methods to estimates the shape parameter and use the estimated value to construct the probability plotting scales.

7 Parametric Likelihood Fitting Concepts: Exponential Distribution

7.1 Introduction

The appeal of maximum likelihood(ML) stems from the fact that it can be applied to a wide variety of statistical models and kinds of data(e.g., continuous, discrete, categorical, censored, truncated).

Statistical theory shows that, under standard regularity conditions, ML estimators are "optimal" in large samples(ML estimators are consistent and asymptotically efficient, i.e., among consistent competitors to ML estimators, non has a smaller asymptotic variance).

Besides Bayesian methods, there is no general theory that suggests alternatives to Ml that will be optimal in finite samples.

Statistical modeling, in practice, is an *iterative* procedure of fitting proposed models in search of a model that provides an adequate description of the population or process of interest without being unnecessarily complicated.

7.2 Parametric Likelihood

Similar with equation (2.6) for non-parametric models, the parametric likelihood function can be written as a joint probability,

$$L(\boldsymbol{\theta}; DATA) = \mathcal{C} \prod_{i=1}^{n} L_i(\boldsymbol{\theta}; data_i). \tag{7.1}$$

In practice, log-likelihood is more widely used with a naturally sums,

$$\mathcal{L}(\boldsymbol{\theta}) = log[L(\boldsymbol{\theta})] = \sum_{i=1}^{n} \mathcal{L}_i(\boldsymbol{\theta}). \tag{7.2}$$

7.3 Confidence Intervals for $\boldsymbol{\theta}$

7.3.1 Likelihood Confidence Intervals for θ

$R(\boldsymbol{\theta}) = \frac{L(\boldsymbol{\theta})}{L(\hat{\boldsymbol{\theta}})}$ is the relative likelihood for $\boldsymbol{\theta}$, then an approximate $100(1-\alpha)\%$ likelihood-based confidence interval is the set of all values of $\boldsymbol{\theta}$

such that

$$-2log(R(\boldsymbol{\theta})) \leq \chi^2_{1-\alpha;n} \tag{7.3}$$

, where n is the degree of number of parameters in $\boldsymbol{\theta}$.

If $-2log(R(\boldsymbol{\theta}_0)) > \chi^2_{1-\alpha;n}$, then the hypothesis testing $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ should be rejected.

7.3.2 Normal-Approximation Confidence Intervals for $\boldsymbol{\theta}$

A $100(1-\alpha)\%$ normal-approximation confidence interval for $\boldsymbol{\theta}$ is

$$[\underset{\sim}{\boldsymbol{\theta}}, \widetilde{\boldsymbol{\theta}}] = \widehat{\boldsymbol{\theta}} \pm z_{(1-\alpha/2)}\hat{se}_{\hat{\boldsymbol{\theta}}}, \tag{7.4}$$

where $\hat{se}_{\hat{\boldsymbol{\theta}}} = \sqrt{\left[-\frac{\mathrm{d}^2\mathcal{L}(\boldsymbol{\theta})}{\mathrm{d}\boldsymbol{\theta}^2}\right]^{-1}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}}$.
It's based on the assumption that $Z = \frac{\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}}{\hat{se}_{\hat{\boldsymbol{\theta}}}}$ can be approximated by NOR$(0,1)$.

An alternative approximate confidence interval for **positive** quantities is,

$$[\underset{\sim}{\theta}, \widetilde{\theta}] = [\hat{\theta}/w, \hat{\theta} \times w], \tag{7.5}$$

where $w = e^{z_{(1-\alpha/2)}\widehat{se}_{\hat{\theta}}/\hat{\theta}}$.

$$\begin{aligned} log(\hat{\theta}) &\pm z_{1-\alpha/2}\widehat{se}_{log(\hat{\theta})} \\ \widehat{se}_{log(\hat{\theta})} &\approx \widehat{se}_{\hat{\theta}}/\hat{\theta} \end{aligned}$$

It's based on the assumption that the distribution

$$Z_{log(\hat{\theta})} = \frac{log(\hat{\theta}) - log(\theta)}{\widehat{se}_{log(\hat{\theta})}} \tag{7.6}$$

can be approximated by a $NOR(0,1)$ distribution.

7.4 Confidence Intervals for Function of θ

For one parameter distributions, confidence intervals for θ can be translated directly into confidence intervals for monotone functions of θ.
For models have more than one parameter, a collection of intervals must be handled differently because the confidence level applies only to the process of constructing an interval for a single point in time t_e.

7.5 Likelihood for Exact Failure Times

7.5.1 Correct Likelihood for Observations Reported as Exact Failures

Sometimes the failure time are reported as exact times, however, are actually discrete. Then the "correct likelihood" can be corrected as

$$L_i(\theta; data_i) = \int_{t_{i-1}}^{ti} f(t;\theta)\mathrm{d}\theta = F(t_i;\theta) - F(t_{i-1};\theta). \tag{7.7}$$

7.5.2 Using Density Approximation for Observations Reported as Exact Values

For most statistical models, the contribution to the likelihood of the observations reported as exact values can, for small $\Delta_i > 0$, be approximated by

$$F(t_i;\theta) - F(t_i - \Delta_i;\theta) \approx f(t_i;\theta)\Delta_i. \tag{7.8}$$

where Δ_i doesn't depend on θ. Then the likelihood

$$L_i(\theta) = f(t_i;\theta) \tag{7.9}$$

differs (7.8) by a constant scale factor. As long as the approximation in (7.8) is adequate, the general character of the likelihood is not affected. (7.9) can be used for the likelihood for an observation with "exact failure time" at t_i.

7.6 Data Analysis with No Failures

It's possible that there will be no failures for data on hight-reliability components. With zero failures and assumed exponential distribution, a conservative $100(1-\alpha)\%$ lower confidence bound on θ is

$$\underset{\sim}{\theta} = \frac{2 \cdot TTT}{\chi^2_{1-\alpha;2}} = \frac{TTT}{-log(\alpha)}, \tag{7.10}$$

while $\chi^2_{1-\alpha;2} = -2log(\alpha)$[2].

[2]Solve the quantile using the chi-square distribution formula, `https://en.wikipedia.org/wiki/Chi-squared_distribution`

8 Maximum Likelihood for Log-Location-Scale Distributions

8.1 Likelihood

$$L(\mu, \sigma) = \prod_{i=1}^{n} [f(y_i; \mu, \sigma)]^{\delta_i} [1 - F(y_i; \mu, \sigma)]^{1-\delta_i}, \tag{8.1}$$

where

$$y = \begin{cases} 1 & \text{if } y_i \text{ is an exact observation} \\ 0 & \text{right-censored observation.} \end{cases}$$

8.2 Likelihood Confidence Regions and Intervals

8.2.1 Joint Confidence Regions for μ and σ

Using the large-sample χ^2 approximation for the distribution of the likelihood ratio statistic, an approximate $100(1-\alpha)\%$ joint confidence region for two dimensional vector,

$$R(\theta_i, \theta_j) > \exp[-\chi^2_{1-\alpha;2}/2]. \tag{8.2}$$

8.2.2 Individual Confidence Intervals for μ and σ

The profile likelihood for μ is

$$R(\mu) = \max_{\sigma} \left[\frac{L(\mu, \sigma)}{L(\hat{\mu}; \hat{\sigma})} \right], \tag{8.3}$$

Given any μ value, one can calculated and find the point of highest relative likelihood by maximizing over σ.
The $100(1-\alpha)\%$ approximate confidence interval interval is given by

$$R(\mu) > \exp[-\chi^2_{1-\alpha;1}/2]. \tag{8.4}$$

Confidence interval for σ can be derived similarly.

8.2.3 Likelihood Confidence Intervals for Functions of μ and σ

Due to the *invariance* property of ML estimators, likelihood based methods can be applied to make inferences about functions of parameters.

For example, for the p quantile, $t_p = \exp[\mu + \Phi^{-1}(p)\sigma]$,

$$R(t_p) = \max_{\sigma} \left[\frac{L(t_p, \sigma)}{L(\hat{\mu}, \hat{\sigma})} \right],$$

where we substitute μ by t_p.

8.3 Normal Approximation Confidence Intervals

1. First compute the variance-covariance matrix. For location-scale distribution, the local estimate $\widehat{\sum}_{\hat{\boldsymbol{\theta}}}$ of $\sum_{\hat{\boldsymbol{\theta}}}$ is the inverse of the observed information matrix,

$$\widehat{\sum}_{\hat{\mu},\hat{\sigma}} = \begin{bmatrix} \widehat{Var}_{\hat{\mu}} & \widehat{Cov}_{\hat{\mu},\hat{\sigma}} \\ \widehat{Cov}_{\hat{\mu},\hat{\sigma}} & \widehat{Var}_{\hat{\sigma}} \end{bmatrix} = \begin{bmatrix} -\frac{\partial^2 \mathcal{L}(\mu,\sigma)}{\partial \mu^2} & -\frac{\partial^2 \mathcal{L}(\mu,\sigma)}{\partial \mu \partial \sigma} \\ -\frac{\partial^2 \mathcal{L}(\mu,\sigma)}{\partial \mu \partial \sigma} & -\frac{\partial^2 \mathcal{L}(\mu,\sigma)}{\partial \mu^2} \end{bmatrix}^{-1}. \tag{8.5}$$

   The partial second derivative describe the curvature of the log likelihood at the ML estimate, More curvature in the log likelihood surface implies a more concentrated likelihood near μ, σ, and this implies more precision.

2. Then for parameters, $\widehat{se}$ can be derived for normal confidence interval approximation.

3. For normal approximation confidence intervals for functions of parameters, say $g = g(\mu, \sigma)$,one can first apply delta method for the variance of the function, i.e.,

$$\widehat{se}_{\hat{g}} = \sqrt{\widehat{Var}(\hat{g})} = \left[\left(\frac{\partial g}{\partial \mu}\right)^2 \widehat{Var}(\hat{\mu}) + 2\left(\frac{\partial g}{\partial \mu}\right)\left(\frac{\partial g}{\partial \sigma}\right) \widehat{Cov}(\hat{\mu}, \hat{\sigma}) + \left(\frac{\partial g}{\partial \sigma}\right)^2 \widehat{Var}(\hat{\sigma}) \right],$$

   then find the approximate confidence interval based on it.

4. The intervals can be improved slightly by using $t_(p, \nu)$ instead of z_p.

8.4 Estimation with Given $\boldsymbol{\sigma}$

With such information available, one can provide considerably more precision from limited data. However, the danger is the given value of σ may be seriously incorrect, resulting in misleading conclusions.

9 Bootstrap Confidence Intervals

Simulation based intervals provide another important method to obtain exact or more accurate approximate confidence intervals. It's expected to be more accurate than the normal approximation methods and competitive with the likelihood-based methods.

9.1 Bootstrap Sampling

General Idea: A confidence interval is judged on the basis of how well the procedure would perform if it were repeated over and over again. A confidence interval should not be too wide and the coverage probability (probability that the interval contains the quantity of interest) should be equal or close to the nominal coverage probability $1 - \alpha$. The **idea** of bootstrap sampling is to simulate the repeated sampling process and use the information from the distribution of appropriate statistics in the bootstrap samples to compute teh needed confidence interval, reducing the reliance on large-sample approximations.

When the goal is to compute confidence intervals, the usual recommendation is to use between $B = 2000$ to $B = 5000$ bootstrap samples. Large B values are recommended for estimating the more extreme quantiles of the bootstrap distribution that are required for higher confidence levels.
Sampling Methods

1. Fully "parametric" bootstrap procedure: Simulate each sample of size n from the assumed parametric distribution,using the ML estimates computed from the actual data to replace the unknown parameters. The disadvantage is it requires complete specification of the censoring process. It's not a problem with simple censoring such as Type I or II , but more difficult for complicated systematic or random censoring.

2. "Non-parametric" bootstrap sampling: each sample of size n is obtained by sampling, with **replacement**, from the actual data cases in the original data set.

9.2 Confidence Intervals

Take exponential distribution as an example.

Monte Carlo simulation-based methods can be used to obtain a better approximation to the distribution of $Z_{log(\hat{\theta})}$ than assuming that $Z_{log(\hat{\theta})} \dot{\sim} NOR(0, 1)$. Here are the steps:

1. Simulate B bootstrap samples of size n;

2. Compute the ML estimates for each bootstrap sample;

3. For the j-th bootstrap sample, compute $Z_{log(\hat{\theta}_j^*)} = \frac{log(\hat{\theta}_j^*) - log(\hat{\theta})}{\widehat{se}_{log(\hat{\theta}_j^*)}}$;
   Intervals are based on the distribution of t-like statistics. , so the method is also called "*boostrap-t*" method.

4. Intervals for θ based on the assumption that the simulation distribution of $Z_{log(\hat{\theta}^*)}$ provides a good approximation of the distribution of $Z_{log(\hat{\theta})}$:

$$[\underset{\sim}{\theta}, \widetilde{\theta}] = [\hat{\theta}/\underset{\sim}{w}, \hat{\theta} \times \widetilde{w}], \tag{9.1}$$

   where $\underset{\sim}{w} = e^{z_{log(\hat{\theta}^*)(1-\alpha/2)} \widehat{se}_{\hat{\theta}}/\hat{\theta}}$, $\widetilde{w} = e^{z_{log(\hat{\theta}^*)(\alpha/2)} \widehat{se}_{\hat{\theta}}/\hat{\theta}}$ and
   $\hat{se}_{\hat{\theta}} = \sqrt{\left[-\frac{\mathrm{d}^2 \mathcal{L}(\theta)}{\mathrm{d}\theta^2} \right]^{-1}_{\theta=\hat{\theta}}}$.
   Note that the w here is not symmetric as Section 7.3.2 and z is based on the *quantile* of the distribution of $Z_{log(\hat{\theta}^*)}$.

5. Intervals for θ based on the assumption that the simulated distribution of $Z_{\hat{\theta}^*}$ provides a good approximation to the distribution of $Z_{\hat{\theta}}$.

$$[\underset{\sim}{\theta}, \widetilde{\theta}] = [\widehat{\theta} - z_{\hat{\theta}^*_{(1-\alpha/2)}} \widehat{se}_{\hat{\theta}}, \quad \widehat{\theta} - z_{\hat{\theta}^*_{(\alpha/2)}} \widehat{se}_{\hat{\theta}}], \tag{9.2}$$

   where the z is the *quantile* of the distribution of $Z_{\hat{\theta}^*}$.

- Bootstrap confidence intervals for monotone functions of θ can be obtained by doing the transformation.

- For distributions with more than one parameter, for example, the Weibull distribution, intervals can be obtained in a similar manner, i.e., base the bootstrap-t confidence intervals on bootstrap evaluations of the distributions of the t-like statistics, for example, $Z_{\hat{\mu}}, Z_{log(\hat{\sigma})}, Z_{logit(\hat{F}(t))} \cdots$

9.3 Percentile Bootstrap Method

When it's not easy to compute the standard error for an estimate, the percentile bootstrap described by Efron and Tibshirani(1993) provides a simple useful alternative.

1. Suppose B simulations , and there is an estimate of θ, $\hat{\theta}_j^*$ for $j = 1, \cdots, B$.

2. The $100(1-\alpha)\%$ percentile bootstrap interval for θ is

$$[\underset{\sim}{\theta}, \widetilde{\theta}] = [\widehat{\theta}^*_{[l]}, \widehat{\theta}^*_{[u]}], \tag{9.3}$$

   where $\widehat{\theta}^*_{[j]}$ is the bootstrap sample ordered from smallest to largest, $l = B \times (\alpha/2)$, $u = B \times (1-\alpha/2)$. And l, u would be rounded to the next lowest and next highest integer, respectively.

The advantage is that the interval doesn't depend the on the transformation scale of the parameter("transformation preerving"). It doens't work as well as the previous bootstrap-t method for small samples, though.

10 Planning Life Test

10.1 Introduction

Idea: When some "planning information" about the life distribution is available, its possible to assess the effect of *sample size* and *test length* on the outcome of a particular test plan. Such planning information is typically obtained from design specifications, expert opinion, or previous experience with similar product or materials.

$\square$ represents a planning value of a population or process quantity.
Simulation of a Proposed Test Plan

1. Use the chosen model and planning values of the distribution to simulate data from the proposed life test.

2. Analyze the data, perhaps fitting more than one distribution.

3. Assess precision of estimates: This can be done initially by computing approximate confidence intervals, as done in the real data.

4. Simulate many samples and fit distributions for each, assess the sample-to-sample differences. Such multiple simulations provide an assessment of estimation precision.

5. Repeat the simulation-evaluation process with different sample sizes to gauge the actual sample size and test length requirements to achieve the desired precision.

6. Repeat the simulation-evaluation process with different input "planning values" over the range of their uncertainty.

In general, simulation is a useful method of assessing variability. To control the standard deviation of an estimator to a specified degree of precision, it's possible to interpolate among simulated values at different sample sizes.

10.2 Approximate Variance of ML Estimators

10.2.1 Basic Large-Sample Approximation

For a model with $\boldsymbol{\theta} = (\theta_i, \ldots, \theta_k)$, the results hold approximately for large samples:

1. $\hat{\boldsymbol{\theta}}$ follow a multivariate normal distribution with mean vector $\boldsymbol{\theta}$ and covariance matrix $\sum_{\hat{\boldsymbol{\theta}}}$.

2. The large sample *approximate* covariance matrix can be computed from $\sum_{\hat{\boldsymbol{\theta}}} = I_{\boldsymbol{\theta}}^{-1}$ with the Fisher information,

$$I_{\boldsymbol{\theta}} = E\Big[-\frac{\partial^2 \mathcal{L}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\Big] = \sum_{i=1}^{n} E\Big[-\frac{\partial^2 \mathcal{L}_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\Big]. \tag{10.1}$$

   In practical problems, interest will center on one or more scalar functions of the parameters, sya $g = g(\boldsymbol{\theta})$. Then in large samples,

I : Assume the distribution of $\hat{g} = g(\hat{\boldsymbol{\theta}})$ can be approximated by a normal distribution,

$$\hat{g} \dot{\sim} \text{NOR}(g(\boldsymbol{\theta}), Ase(\hat{g}), \tag{10.2}$$

where

$$\begin{aligned} Ase(\hat{g}) &= \frac{1}{\sqrt{n}} \sqrt{V_{\hat{g}}} \\ V(\hat{g}) &= nAvar(\hat{g}) \\ Avar(\hat{g}) &= \Big[\frac{\partial g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big]' \sum\nolimits_{\hat{\boldsymbol{\theta}}} \Big[\frac{\partial g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big] \end{aligned}$$

II : If $g(\boldsymbol{\theta})$ is positive for all $\boldsymbol{\theta}$, then an alternate form is better. Assume the distribution of $log(\hat{g}) = log(g(\hat{\boldsymbol{\theta}}))$ can be approximated by a normal distribution,

$$log(\hat{g}) \dot{\sim} \text{NOR}(log(g(\boldsymbol{\theta})), Ase(log(\hat{g})), \tag{10.3}$$

where

$$\begin{aligned} Ase(log(\hat{g})) &= \frac{1}{\sqrt{n}} \sqrt{V_{log(\hat{g})}} \\ V(log(\hat{g})) &= nAvar(log(\hat{g})) \\ Avar(log(\hat{g})) &= \Big(\frac{1}{g}\Big)^2 Avar(\hat{g}) \end{aligned}$$

10.3 Sample Size for Unrestricted Functions

Assume normal distribution of $\hat{g}$, and the actual confidence interval*half-width* is D,

$$D = z_{(1-\alpha/2)} \frac{1}{\sqrt{n}} \sqrt{\widehat{V}_{\hat{g}}} \tag{10.4}$$

$$n = \frac{z^2_{(1-\alpha/2)} V^{\square}_{\hat{g}}}{D^2}. \tag{10.5}$$

Similar way is used for sample sizes for other kinds of distributions.

11 Parametric Maximum Likelihood: Other Models

11.1 Truncated Data and Truncated Distributions

Censored Data	Truncated Data
It occurs when there is a bound on an observation, that is, lower bounds for observations censored on the right, upper bounds for obs censored on the left, and both upper and lower bounds for interval censored data.	It arises when even the existence of a potential observation would be unknown, if its value were to lie in a certain range.

Table 1: Difference between Censored and Truncated Data

1. Likelihood with Left Truncation:

$$L_i(\boldsymbol{\theta}) = \mathrm{P}(t_i^L < T_i \leq t_i | T_i > \tau_i^L) = \frac{F(t_i;\boldsymbol{\theta}) - F(t_i^L;\boldsymbol{\theta})}{1 - F(\tau_i^L;\boldsymbol{\theta})}, \quad t_i > t_i^L \geq \tau_i^L. \tag{11.1}$$

   For an observation with exact failure time t_i,

$$L_i(\boldsymbol{\theta}) = \frac{f(t_i;\boldsymbol{\theta})}{1 - F(\tau_i^L;\boldsymbol{\theta})}, \quad t_i > \tau_i^L. \tag{11.2}$$

2. Likelihood with Right Truncation: similar with the left truncation.

3. Likelihood with Right and Left Truncation for an interval censored observation:

$$L_i(\boldsymbol{\theta}) = \mathrm{P}(t_i^L < T_i \leq t_i | \tau^L \leq T < \tau^U) = \frac{F(t_i;\boldsymbol{\theta}) - F(t_i^L;\boldsymbol{\theta})}{F(\tau_i^U;\boldsymbol{\theta}) - F(\tau_i^L;\boldsymbol{\theta})}, \quad \tau_i^L \leq t_i^L < t_i \leq \tau_i^U. \tag{11.3}$$

11.2 Summary

General guidelines for fitting parametric distributions to skewed data.

1. If data is left-skewed, it's possible to fit a three-parameter Weibull distribution and achieve a good fit to the data. In many cases, however, it will be possible to fit the simpler two-parameter Weibull or smallest extreme value distributions and get, effectively, the same results.

2. If the data is approximately symmetric, one can generally fit a three-parameter Weibull, a three-parameter lognormal model, or two-parameter versions of the distributions and get a reasonably good fit. In many cases, however, it will be possible to fit the simpler two-parameter normal or logistic distributions and get, effectively, the same results.

3. If the data is right skewed, it's often to fit a three-parameter Weibull or lognormal distribution to get a good fit of the data.

4. The use of a threshold parameter can be viewed from two different directions. Sometimes it might be viewed as a physical parameter — a time before which probability of failure is zero or a threshold strength. In other cases, γ is one of several parameters of a curve being fit to the data.

12 Prediction of Future Random Quantiles

12.1 Probability Prediction Intervals($\boldsymbol{\theta}$ Given)

With a completely specified continuous probability distribution, an exact $100(1-\alpha)\%$ "probability prediction interval" for a further observation from $F(t;\boldsymbol{\theta})$ is (ignoring the data),

$$PI(1-\alpha) = [\underset{\sim}{T}, \widetilde{T}] = [t_{\alpha/2}, t_{1-\alpha/2}], \tag{12.1}$$

where t_p is the p quantile of $F(t;\boldsymbol{\theta})$. The coverage probability of the interval in (12.1) is,

$$P[T \in PI(1-\alpha)] = P[\underset{\sim}{T} \leq T \leq \widetilde{T}] = P[t_{\alpha/2} \leq T \leq t_{1-\alpha/2}] = 1-\alpha \quad (12.2)$$

for a continuous distribution.

12.2 Statistical Prediction Interval($\boldsymbol{\theta}$ Estimated)

12.2.1 Coverage Probability Concepts

In statistical prediction, the objective is to predict the random quantity T based on "learning" sample information(denoted by $DATA$). Generally with only sample data, there is uncertainty in the distribution parameters. There are two kinds of coverage probabilities:

1. For fixed DATA (and thus fixed $\hat{\boldsymbol{\theta}}$ and $[\underset{\sim}{T}, \widetilde{T}]$), the conditional coverage probability of a particular interval $[\underset{\sim}{T}, \widetilde{T}]$ is,

   $$CP[PI(1-\alpha)|\hat{\boldsymbol{\theta}};\boldsymbol{\theta}] = P(\underset{\sim}{T} \leq T \leq \widetilde{T}|\hat{\boldsymbol{\theta}};\boldsymbol{\theta}) = F(\widetilde{T};\boldsymbol{\theta}) - F(\underset{\sim}{T};\boldsymbol{\theta}). \quad (12.3)$$

   This conditional probability is *unknown* because $F(t;\boldsymbol{\theta})$ depends on the unknown $\boldsymbol{\theta}$.

2. From sample to sample, the conditional coverage probability is *random* because $[\underset{\sim}{T}, \widetilde{T}]$ depends on $\hat{\boldsymbol{\theta}}$. The unconditional coverage probability for the prediction interval *procedure* is

   $$CP[PI(1-\alpha)|\boldsymbol{\theta}] = P(\underset{\sim}{T} \leq T \leq \widetilde{T}|\boldsymbol{\theta}) = E_{\hat{\boldsymbol{\theta}}}\{CP[PI(1-\alpha)|\hat{\boldsymbol{\theta}},\boldsymbol{\theta}]\}. \quad (12.4)$$

   This is the one generally used to describe a prediction interval procedure. In general, $CP[PI(1-\alpha);\boldsymbol{\theta}] \approx 1-\alpha$ because of the dependency on the unknown $\boldsymbol{\theta}$.

12.2.2 Naive Method for Computing a Statistical Prediction Interval

A "naive" prediction interval for continuous T is obtained by substituting the maximum likelihood estimate for $\boldsymbol{\theta}$ into (12.1).

12.3 The (Approximate) Pivotal Method for Prediction Intervals

12.3.1 Type II (Failure) Censoring

A life test is run until a specified number of failures (r). When T has a log-location-scale distribution, and the data are complete or Type II censored, the random variable $Z_{log(T)}$ is pivotal,i.e., the distribution depends only on n and r, not $\boldsymbol{\theta}$.
The prediction interval can be obtained from,

$$P[\hat{\mu} + z_{log(T)_{\alpha/2}} \times \hat{\sigma} < log(T) \leq \hat{\mu} + z_{log(T)_{1-\alpha/2}} \times \hat{\sigma}] = 1 - \alpha. \tag{12.5}$$

Then quantiles $z_{log(T)_{\alpha/2}}$, $z_{log(T)_{1-\alpha/2}}$ can be obtained from the distribution of $Z_{log(T)}$, which can be obtained approximately(due only to Monte Carlo error) by simulating B realizations of $Z_{log(T^*)} = \frac{log(T^*)-\hat{\mu}^*}{\hat{\sigma}^*}$ with the following steps,

1. Draw a sample of size n from a log-location-scale distribution with parameters $(\hat{\mu}, \hat{\sigma})$, censored at the rth failure.
2. Use the censored sample to compute ML estimates $\hat{\mu}^*$ and $\hat{\sigma}^*$.
3. Draw an additional single observation T^* from the log-location-scale distribution with parameters $(\hat{\mu}, \hat{\sigma})$.
4. Compute $Z_{log(T^*)} = \frac{log(T^*)-\hat{\mu}^*}{\hat{\sigma}^*}$.
5. Repeat 1 to 4 B times. Obtain the approximation for the quantiles $z_{log(T)_{\alpha/2}}$ and $z_{log(T)_{1-\alpha/2}}$ from the empirical distribution of $Z_{log(T^*)}$.

12.3.2 Type I Censoring

$Z_{log(T)}$ is only approximately pivotal and quantiles of $Z_{log(T)}$ depend on $F(t_c; \boldsymbol{\theta})$, the unknown expected proportion failing by time t_c, and the sample size n. Thus the prediction interval to predict T is approximate.

13 Degradation Data, Models, and Data Analysis

13.1 Introduction

Design of high-reliability systems generally requires that the individual system components have extremely high reliability, even after a long

periods of time. With short development times, tests must be conducted with severe time constrains. Frequently no failures occur during such tests. A relationship between component failure and amount of degradation makes it possible to use degradation models and data to make inferences and predictions about failure time.

13.2 Models for Degradation Data

13.2.1 Degradation Data

Degradation Data:

1. The measurement of physical degradation as a function of time.
2. Actual physical degradation cannot be measured, but measures of product performance degradation may be available.

Both kinds are generally referred to as "degradation data". Modeling performance degradation data may be useful but complicated because performance may be affected by more than one underlying degradation process.

Advantages of degradation data:

1. Degradation data can provide considerably more reliability information than traditional censored failure-time data.
2. Accelerated tests are commonly used obtain reliability tests information more quickly. Direct observation of the physical degradation process or some closely related surrogate may allow direct modeling of the failure causing mechanism, providing more credible and precise reliability estimates and a firmer basis for often-needed extrapolation.

13.2.2 Degradation Leading to Failure

Most failures can be traced to an underlying degradation process. Figure 2 on Page 40 shows different possible shapes of the degradation curves.

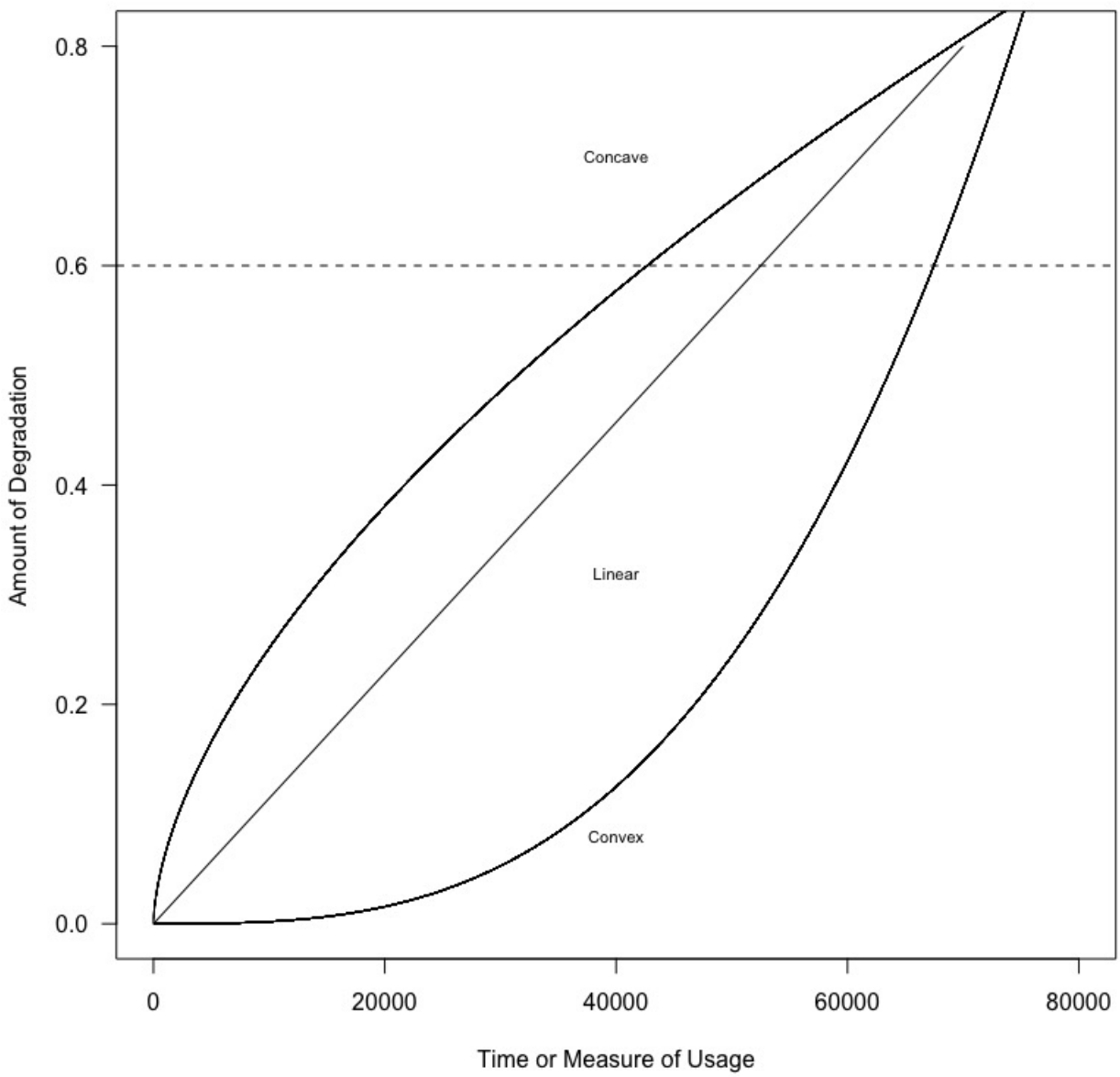

Figure 2: Possible Shapes for Univariate Degradation Curves

13.2.3 Models for Variation in Degradation and Failure Times

There is some degree of variability in all of model factors as well as in factors that are not in the model. These factors combine to cause variability in the degradation curves and in failure times.

1. Unit-to-Unit Variability
   - Initial Conditions
   - Material Properties
   - Component Geometry or Dimensions
   - Within Unit Variability: often there will be spatial variability in material properties within a unit(e.g., defects).
2. Variability Due to Operating and Environmental Conditions: For example, stress and temperature.

13.2.4 Limitations of Degradation Data

1. Physical degradation or performance degradation are natural properties to measure for many testing process. However, degradation measurements, often, requires destructive inspection or disruptive measurement. In such situations, one can obtain only one single measurement on each unit tested.
2. Also, when degradation data are contaminated with large amounts of measurement error or when the degradation measure is not closely related to a failure, the advantages of degradation data is compromised. For example, when the measurement is on performance degradation, failures may occur for physical reasons that cannot be observed directly.

An important but difficult engineering challenge of degradation analysis is to find variables that are closely related to failure time and develop methods for accurately measuring these variables.

13.2.5 General Degradation Path Model

The actual degradation data of a particular unit over time is denoted by $\mathcal{D}(t), t > 0$. The observed sample degradation $y_i j$ of unit i and time t_j,

$$y_{ij} = \mathcal{D}_{ij} + \epsilon_{ij}, \quad i = 1, \ldots, n, \quad j = 1, \ldots, m_i, \tag{13.1}$$

where $\mathcal{D}_{ij} = \mathcal{D}(t_{ij}, \beta_{1i}, \ldots, \beta_{ki})$ is the actual path of the unit and $\epsilon_{ij} \sim NOR(0, \sigma_\epsilon)$. Typically ,$\boldsymbol{\beta}$ is a vector of k unknown parameters, same paths have $k = 1, 2, 3, 4$ parameters.Some $\boldsymbol{\beta}$ will be **random** from unit to unit.
Assumptions:

1. Random $\boldsymbol{\beta}$ is independent of the ϵ_{ij} deviations.
2. σ_ϵ is constant. The adequacy of this assumption can be affected by transforming $\mathcal{D}(t)$. If there is an autocorrelation among $\epsilon_{ij}, j = 1, \ldots, m_i$, one can use a time series model for the residual term along with appropriate estimation methods.

13.2.6 Degradation Model Parameters

We focus on making inferences about the population or process or predictions about future units. The underlying model parameters are $\boldsymbol{\theta}_{\boldsymbol{\beta}} = (\boldsymbol{\mu}_{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}})$, which is the mean and covariance matrix of random $\boldsymbol{\beta}$.

13.3 Estimation of Degradation Model Parameters

The likelihood for the *random-parameter* degradation model in Section 13.2.5,

$$\begin{aligned} L(\boldsymbol{\theta}_{\boldsymbol{\beta}}, \sigma_\epsilon | DATA) &= \prod_{i=1}^{n} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left[\prod_{j=1}^{m_i} \frac{1}{\sigma_\epsilon} \phi_{nor}(\zeta_{ij}) \right] \times \\ & f_{\boldsymbol{\beta}}(\beta_{1i}, \ldots, \beta_{ki}; \boldsymbol{\theta}_{\boldsymbol{\beta}}) \mathrm{d}\beta_{1i}, \ldots, \mathrm{d}\beta_{ki}, \end{aligned} \tag{13.2}$$

where $\zeta_{ij} = \frac{y_{ij} - \mathcal{D}(t_i j, \beta_{1i}, \ldots, \beta_{ki})}{\sigma_\epsilon}$ and $f_{\boldsymbol{\beta}}(\beta_{1i}, \ldots, \beta_{ki}; \boldsymbol{\theta}_{\boldsymbol{\beta}})$ is the multivariate normal distribution density function.

Parameters needed to be estimated are $(\boldsymbol{\mu}_{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}}, \sigma_\epsilon)$.

13.4 Models Relating Degradation and Failure

13.4.1 Soft Failures: Specified Degradation Level

The failure is defined (in a somewhat arbitrary, but purposeful, manner) at a specified level of degradation if the products have a gradual loss of performance. For example, light bulb output.

A fixed $\mathcal{D}_f$, the critical degradation level for failure, will be used.

13.4.2 Hard Failures: Joint Distribution of Degradation and Failure Level

The definition of a failure is clear — the product stops working.

With hard failures, failure times will not, in general, correspond exactly with a particular level of degradation. Instead, the level of degradation at which failure occurs will be random from unit to unit and even over time. This could be modeled by using a distribution to describe unit to unit variability in $\mathcal{D}_f$ or a joint distribution of $\boldsymbol{\beta}$ and stochastic behavior in $\mathcal{D}_f$, where $\mathcal{D}_f$ is the critical degradation level for failure.

13.5 Evaluation of $F(t)$

A specified model for $\mathcal{D}_t$ and $\mathcal{D}_f$ defines a failure-time distribution. This distribution can be written as a function of the degradation model parameters. Suppose that if the degradation level first reaches $\mathcal{D}_f$ at time t, then the unit fails at time t. Then

$$P(T \leq t) = F(t) = F(t; \boldsymbol{\theta_\beta}) = P(\mathcal{D}(t; \beta_1, \ldots, \beta_k) \geq \mathcal{D}_f). \tag{13.3}$$

That is to say, for a fixed $\mathcal{D}_f$, the distribution of T depends on the distribution of the $\beta_1, \ldots, \beta_k$, which in turn, depends on the basic path parameters $\boldsymbol{\theta_\beta}$.
For most practical models, especially when $\mathcal{D}(t)$ is nonlinear and more than one $\beta_1, \ldots, \beta_k$ are random, $F(t)$ may not be written in a closed form and numerical evaluation methods need to be used.

13.5.1 Analytic Solution for $F(t)$

For some *simple* path models, $F(t)$ can be expressed as a function of the basic path parameters of a particular unit,

$$\mathcal{D}(t) = \beta_1 + \beta_2 t \tag{13.4}$$

, where β_1 is fixed and $\beta_2 \sim LOGNOR(\mu, \sigma)$, i.e., β_1 is the common initial amount of degradation of all the test units, and β_2 is the random degradation rates for different units. Then the $F(t)$,

$$\begin{aligned} F(t; \beta_1, \mu, \sigma) &= P[\mathcal{D}(t) \geq \mathcal{D}_f] \\ &= P[\beta_2 > \frac{\mathcal{D}_f - \beta_1}{t}] \\ &= 1 - \Phi_{nor}\Big(\frac{log(\frac{\mathcal{D}_f - \beta_1}{t}) - \mu}{\sigma}\Big), \quad t > 0 \end{aligned} \tag{13.5}$$

13.5.2 Numerical Evaluation of $F(t)$

When a closed form of $F(t)$ exists, one can evaluate it by direct integration.

13.5.3 Monte Carlo Evaluation of $F(t)$

Monte Carlo simulation is a versatile method for evaluating $F(t)$. The idea is: generate a large number of random sample paths from the assumed path model, using the proportion of path crossing $\mathcal{D}_f$ by time t as an evaluation of $F(t)$.

Algorithm

1. Generate N simulated realizations of $\check{\beta}_1, \ldots, \check{\beta}_k$ of $\beta_1, \ldots, \beta_k$ from a multivariate normal distribution with mean $\hat{\boldsymbol{\mu}}_{\boldsymbol{\beta}}$ and covariance matrix $\hat{\Sigma_{\boldsymbol{\beta}}}$. N is a large number, for example, 10,000.

2. Compute N simulated failure times corresponding to the N realizations of $\check{\beta}_1, \ldots, \check{\beta}_k$ by solving $\mathcal{D}(t; \beta_{1i}, \ldots, \beta_{ki}) = \mathcal{D}_f$.

3. For any desired values of t, compute

$$F(t) \approx \frac{Number\, of\, Simulated\, First\, Crossing\, Time \leq t}{N}, \tag{13.6}$$

   and this is an evaluation of $F(t)$.

The error of this approximation is evaluated by using the binomial distribution: The standard deviation of the Monte Carlo error in $F(t)$ is $\sqrt{F(t)(1 - F(t))/N}$.

13.6 Estimation of $F(t)$

1. Closed form exists: The failure time distribution $F(t)$ by substituting the estimates $\hat{\boldsymbol{\theta}}_{\boldsymbol{\beta}}$ into Equation (13.3).

2. No closed form: Algorithm in Section 13.5.2 and 13.5.3 can be used to evaluate equation (13.3) at $\hat{\boldsymbol{\theta}_{\boldsymbol{\beta}}}$.

13.7 Bootstrap Confidence Intervals

Bias-corrected percentile bootstrap method is used for obtain standard errors for $\hat{F}(t)$.

1. Use the observed data from the n sample paths to compute the estimates $\hat{\boldsymbol{\theta}}_{\boldsymbol{\beta}}$ and $\hat{\sigma}_{\epsilon}$.

2. Use the algorithm in Section 13.5.2 and 13.5.3(either numeric or Monte Carlo evaluation) to compute the estimate $\hat{F}(t)$ at desired values of t. This the estimated $F(t)$.

3. Generate a large number B(e.g. $B = 4000$) of bootstrap samples that *mimic* the original sample and compute the corresponding bootstrap estimates $\hat{F}^*(t)$ with the following steps:

   - Generate n simulated realizations of the random path parameters $\beta^*_{1i}, \ldots, \beta^*_{ki}$, $i = 1, \ldots, n$ from $\hat{\boldsymbol{\theta}}_{\boldsymbol{\beta}}$.
   - Using the same sampling scheme as in the *original* experiment, compute n simulated observed paths from Equation (13.1)

   $$y^*_{ij} = \mathcal{D}^*_{ij} + \epsilon^*_{ij}, \quad i = 1, \ldots, n, \quad j = 1, \ldots, m_i, \tag{13.7}$$

   up to the planned stopping time t_c, where ϵ^*_{ij} is from the normal distribution based $\hat{\sigma}_\epsilon$ in Step 1, and $\mathcal{D}^*_{ij} = \mathcal{D}(t_{ij}; \beta^*_{1i}, \ldots, \beta^*_{ki})$, t_{ij} is the same as the original.
   - Use the n simulated paths to estimate parameters of the path model, giving the bootstrap estimates $\hat{\boldsymbol{\theta}}^*_{\boldsymbol{\beta}}$.
   - Use the algorithm in Section 13.5.2 and 13.5.3(either numeric or Monte Carlo evaluation) to compute the estimate $\hat{F}^*(t)$ at desired values of t.

4. For each desired value of t, the bootstrap confidence interval for $F(t)$ is calculated using the following steps with **bias-corrected percentile bootstrap**:

   - Sort the B estimates $\hat{F}^*(t)_1, \ldots, \hat{F}^*(t)_B$ in increasing order.
   - The lower and upper bounds of point-wise approximate $100(1-\alpha)\%$ confidence intervals for the distribution function $F(t)$ according to [Tibshirani(1993)]:

   $$\left[\underset{\sim}{F}(t), \widetilde{F}(t)\right] = \left[\hat{F}^*(t)_{(l)}, \hat{F}^*(t)_{(u)}\right], \tag{13.8}$$

   where

   $$\begin{aligned} l &= B \times \Phi_{nor}\left[2\Phi^{-1}_{nor}(q) + \Phi^{-1}_{nor}(\alpha/2)\right] \\ u &= B \times \Phi_{nor}\left[2\Phi^{-1}_{nor}(q) + \Phi^{-1}_{nor}(1-\alpha/2)\right], \end{aligned}$$

   and q is the proportion of the B values of $\hat{F}^*(t)$ that are less than $\hat{F}(t)$. $q = 0.5$ is equivalent to the percentile bootstrap.

13.8 Comparison with Traditional Failure-Time Analysis

Degradation directly models the relationship degradation and time and takes account of the censored observation when estimating $F(t)$, so sometimes it can provide a more reasonable extrapolation on $F(t)$ than the traditional failure time analysis.

13.9 Approximate Degradation Analysis

Approximate degradation analysis provides an (approximately correct) alternative method of analyzing degradation data.
Steps:

1. First step: separate analysis for each unit to predict the time at which the unit will reach the critical degradation level corresponding to failure.
   - For the unit i, use the path model $y_{ij} = \mathcal{D}_{ij} + \epsilon_{ij}$ and the sample path data $(t_{i1}, y_{i1}), \ldots, (t_{ik}, y_{ik})$ to find the conditional ML estimates $\hat{\boldsymbol{\beta}}_i = (\hat{\beta}_{1i}, \ldots, \hat{\beta}_{ki})$.
   - Solve $\mathcal{D}(t; \hat{\boldsymbol{\beta}}_i) = \mathcal{D}_f$ for t and call the solution $\hat{t}_i$, i.e., the estimated failure time for unit i.
2. Second step: use all the pseudo failure times from Step 1 as a complete sample of failure times to estimate $F(t)$: Repeat the procedure for each sample path to obtain all n pseudo failure times. Do a single distribution analysis of data $\hat{t}_1, \ldots, \hat{t}_n$.

Potential problems:

- Less appealing when the degradation paths are nonlinear.
- It ignores the prediction error in $\hat{t}$ and doesn't account for the measurement error in the observed sample paths.
- The distribution fitted to the pseudo failure times will not, in general, correspond to the distribution induced by the degradation model.
- For some applications, the sample paths may no contain enough information to estimate all parameters at some certain time. Fitting different models for different sample paths could be necessitated.

Overall, extrapolation into the tails of the failure time distribution may be more valid with the actual crossing distribution implied by the degradation model than with the empirically predicted failure times.

14 Introduction to the Use of Bayesian Methods for Reliability Data

14.1 Introduction

Combination of extensive past experience and physical/chemical theory can provide prior information to form a framework for inference and decision making. However, there are, of course, dangers involved in making strong assumptions about knowledge of model parameters.

14.2 Using Bayes Rule to Update Prior Information

The posterior distribution of a set of parameters,

$$f(\boldsymbol{\theta}|DATA) = \frac{L(\text{DATA}|\boldsymbol{\theta})f(\boldsymbol{\theta})}{\int L(\text{DATA}|\boldsymbol{\theta})f(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}} = \frac{R(\boldsymbol{\theta})f(\boldsymbol{\theta})}{\int R(\boldsymbol{\theta})f(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}}, \tag{14.1}$$

where $R(\boldsymbol{\theta}) = \frac{L(\boldsymbol{\theta})}{L(\hat{\boldsymbol{\theta}})}$ is the relative likelihood.
There is usually no closed form of this posterior distribution.

14.3 Prior Information and Distributions

Two main sources of prior information:

1. Expert or other subjective opinion.
2. Past data.

14.3.1 Noninformative(diffuse) Prior Distributions

Non-informative pdfs that are constant over the range of the model parameters. It's also called "vague prior" or "diffuse prior".

14.3.2 Using Past Data to Specify a Prior Distribution

Combining past data with a non-informative prior distribution gives a posterior pdf that is proportional to the likelihood.

14.3.3 Expert Opinion and Eliciting Prior Information

A general approach is to elicit information about particular quantities(or parameters) that, from past experience(or data), can be specified approximately independently. For example, for a high reliability integrated circuit, a good choice would be a quantile in the lower tail of the failure-time distribution and the lognormal shape parameter σ.

14.4 Numerical Methods for Combining Prior Information with a Likelihood

14.4.1 Simulation-based Methods for Computing the Posterior Distribution of $\boldsymbol{\theta}$

Using a larger number of simulated points provides a better approximation and the number of points used is limited only by computing equipment and time constrains.

Algorithm 14.1 Monte Carlo Simulation

1. Generate a random sample , $\boldsymbol{\theta}_i$, $i = 1, \ldots, M$, from the prior $f(\boldsymbol{\theta})$.
2. Retain the ith sample $\boldsymbol{\theta}_i$, with probability $R(\boldsymbol{\theta}_i)$, the relative likelihood. Do this by generating a random variable U_i from uniform distribution and retain the sample if $U_i \leq R(\boldsymbol{\theta}_i)$.

14.4.2 Marginal Posterior Distributions

Inferences on **individual** parameters are obtained by using the marginal posterior distribution of the parameter of interest.

$$f(\theta_j|DATA) = \int f(\boldsymbol{\theta}|DATA)\mathrm{d}\boldsymbol{\theta}_{-j} \tag{14.2}$$

Estimates and confidence intervals for a scalar function of the parameters $g(\boldsymbol{\theta})$ are obtained by using the marginal posterior pdf $f[g(\boldsymbol{\theta}|DATA)]$ and cdf $F[g(\boldsymbol{\theta}|DATA)]$, which are approximated by the empirical pdf and cdf of $g(\boldsymbol{\theta}^*)$ by simulation, respectively.

14.5 Using The Posterior Distribution for Estimation

14.5.1 Bayesian Point Estimation

Bayesian inference for $\boldsymbol{\theta}$ and functions of the parameters $g(\boldsymbol{\theta})$ are entirely based on, $f[\boldsymbol{\theta}|DATA]$ and $f[g(\boldsymbol{\theta}|DATA)]$ respectively. Given that $g(\boldsymbol{\theta})$ is a

scalar, a common Bayesian estimate of $g(\boldsymbol{\theta})$ is the mean of the posterior distribution.

$$\hat{g}(\boldsymbol{\theta}) = E[g(\boldsymbol{\theta})|DATA] = \int g(\boldsymbol{\theta}) f(\boldsymbol{\theta}|DATA)\mathrm{d}\boldsymbol{\theta}. \tag{14.3}$$

This is an estimate that minimizes the square error loss. Other possible choices to estimate $g(\boldsymbol{\theta})$ are mode of the posterior pdf(similar to the ML estimate), median and mean. For example,

$$\hat{g}(\boldsymbol{\theta}) \approx \frac{1}{M^*} \sum_{i=1}^{M^*} g(\boldsymbol{\theta}_i^*),$$

is the sample mean.

14.5.2 Bayesian Interval Estimation

A $100(1-\alpha)\%$ confidence interval(credible interval) for a scalar function $g(\boldsymbol{\theta})$ is any interval satisfying

$$\int_{\underset{\sim}{g}}^{\widetilde{g}} f[g(\boldsymbol{\theta})|DATA]\mathrm{d}g(\boldsymbol{\theta}) = 1-\alpha. \tag{14.4}$$

The lower and upper bounds can be chosen in different ways:

1. Combine two $100(1-\alpha/2)\%$ intervals: it puts equal probability in each tail, preferred when there is more concern for being incorrect in one direction than the other.

2. Highest posterior density(HPD): it chooses $[\underset{\sim}{g}, \widehat{g}]$ which consists all values of g with $f(g|DATA) > c$, where c is a constant such that (14.4) holds. It's similar to likelihood-based approximate confidence intervals, calibrated with a χ^2 quantile.

   If there are more than one parameters in $\boldsymbol{\theta}$, a $100(1-\alpha)\%$ confidence region will be computed:

   $$CR_B = \{g(\boldsymbol{\theta})|f[g(\boldsymbol{\theta})|DATA] \geq c\}, \tag{14.5}$$

   where c is chosen such that $\int_{CR_B} f[g(\boldsymbol{\theta})|DATA]\mathrm{d}g(\boldsymbol{\theta}) = 1-\alpha$.

14.6 Bayesian Prediction

Bayesian methods are useful for predicting future event like the failure of a unit from a specified population or process. It can be predicted by using the Bayesian posterior predictive distribution.

14.6.1 Bayesian Posterior Predictive Distribution

If X represents a future random variable from $f(x|\boldsymbol{\theta})$, then the posterior predictive pdf of X:

$$f(x|DATA) = \int f(x|\boldsymbol{\theta})f(\boldsymbol{\theta}|DATA)\mathrm{d}\boldsymbol{\theta} = E_{\boldsymbol{\theta}|DATA}[f(x|\boldsymbol{\theta})]. \tag{14.6}$$

14.6.2 Approximating Posterior Predictive Distribution

The Bayesian posterior predictive pdf can be approximated by the average of the posterior pdf $f(x|\boldsymbol{\theta}_i^*)$,

$$f(x|DATA) \approx \frac{1}{M^*}\sum_{i=1}^{M^*} f(x|\boldsymbol{\theta}_i^*), \tag{14.7}$$

where $\boldsymbol{\theta}_i^*$ is sampled using Algorithm 14.1. The CDF can be approximated similarly by replacing f with F.

14.6.3 Posterior Predictive Distribution for the kth Failure from a Future Sample of Size m

Assume distribution of time T has a log-location-scale distribution. Let $T_{(k)}$ denote the kth largest observation for a sample of size m from the distribution of T. The pdf of $T_{(k)}$ conditional on $\boldsymbol{\theta}$,

$$f(t_{(k)}|\boldsymbol{\theta}) = \frac{m!}{(k-1)!(m-k)!} \times [\Phi(\zeta)]^{k-1} \times \frac{1}{\sigma t_{(k)}}\phi(\zeta) \times [1-\Phi(\zeta)]^{m-k}, \tag{14.8}$$

where $\zeta = \frac{log(t_{(k)})-\mu}{\sigma}$.
The CDF of $T_{(k)}$ conditional on $\boldsymbol{\theta}$ is,

$$P[T_{(k)} \leq t_{(k)}|\boldsymbol{\theta}] = F[t_{(k)}|\boldsymbol{\theta}] = \sum_{j=k}^{m} \frac{m!}{j!(m-j)!}[\Phi(\zeta)]^j \times [1-\Phi(\zeta)]^{m-j}. \tag{14.9}$$

Combining (14.7), (14.8) and (14.9) provides the pdf and cdf of $T_{(k)}$.

14.7 Practical Issues in the Application of Bayesian Methods

14.7.1 Comparison Between Bayesian and Likelihood/Frequentist Statistical Methods

Difference: The manner in which nuisance parameters are handled.

Bayesian	Likelihood
Bayesian interval inference methods are based on a marginal distribtion in which nuisance parameters have been integrated out.	Nuisance parameters can be maximized out, as suggested by large-sample theory.
Parameter uncertainty can be interpreted in terms of probabilities from the marginal posterior distribution.	Confidence intervals based on likelihood and profile likelihood functions can be calibrated and interpreted in terms of repeated sampling coverage probabilities.
In large samples where the likelihood and posterior are approximately symmetric, Bayesian and likelihood confidence interval methods give very similar answers when prior information is approximately uninformative.	

Table 2: Comparison of Bayesian and Likelihood Methods

14.7.2 Caution on the Use of Prior Information

Analysts and decision makers must beware of and avoid the use of "wishful thinking" as prior information. The potential for generating seriously misleading conclusions is especially high when experimental data will be limited and the prior distribution will dominate in the final answers.

When using Bayesian statistics, it's important to do **sensitivity analysis** with respect to uncertain inputs to one's model, i.e., change the prior distribution assumptions and check the effect that the changes have on final answers of interest.

15 System Reliability Concepts and Methods

15.1 Introduction

Assessing and improving system reliability generally requires consideration of system structure and component reliability. Also, some systems are replaced upon failure, while many are maintained and/or repaired after failure. For repairable systems, availability may be the appropriate metric. This leads to consideration of maintainability and repairability. In general, availability can be increased by increasing reliability or by improving

maintainability and repairability.

- Availability: the fraction of time taht a system is available for use.
- Maintainability: improvement of reliability through inspection and/or preventive maintenance.
- Repairability: characterized by the distribution of time to do a repair.

15.2 System Structures and System Failure Probability

System failure probability, $F_T(t; \boldsymbol{\theta})$, is the probability that the system fails before t. The failure probability of the system is a function of

- time in operation,
- the system structure,
- reliability of system components,
- interconnections, and interfaces.

Complicated system structures can generally be decomposed into collections of the simpler structures, and the methods for evaluation of system reliability can be adapted to more complicated structures.

15.2.1 Time Dependency of System Reliability

Time dependency of the survival probability is suppressed in this Chapter. Then the cdf of a system with s components is
$F_T(\boldsymbol{\theta}) = g[F_1(\boldsymbol{\theta}_1), \ldots, F_s(\boldsymbol{\theta}_s)]$.

15.2.2 System with Component in Series

For a system with s independent components, the cdf is

$$F_T(t) = 1 - \prod_{i=1}^{s}(1 - F_i)). \tag{15.1}$$

The system hazard function is the sum of component hazard functions,

$$h_T(t) = \sum_{i=1}^{s} h_i(t), \tag{15.2}$$

which can be derived by hazard function definition in Chapter 2:

$$\begin{aligned} F(t) &= 1 - e^{-H(t)} = \prod_{i=1}^{s}(1 - F_i) \Rightarrow \\ H(t) &= -log(1 - F(t)) = -\sum_{i=1}^{s} log(1 - F_i) = \sum H_i. \end{aligned}$$

Importance of Part Count in Product Design:
Rule of thumb in reliability engineering:keep the part count small- keep the number of individual components in a system to a minimum.

Effect of Positive Dependency in a Two-Component System:
The assumption of independence is *conservative* in the sense that the actual $F_T(t)$ is smaller than that predicted by the independent component model. If the correlation is negative, the prediction will be anti-conservative. But negative correlation is uncommon in physical systems.

15.2.3 System with Components in Parallel

For a system with s independent components, the cdf of the system is :

$$F_T(t) = \prod_{i=1}^{s} F_i. \tag{15.3}$$

Effect of Positive Dependency in a Two-Component Parallel-Redundant System:
The advantage of redundancy can be degraded seriously when the failure times of the individual components have positive dependence, as the system reliability will be smaller if there exits a positive relationship.

15.2.4 Systems with Components in Series-Parallel

Series-Parallel System Structure with System-Level Redundancy
A $r \times k$ series-parallel system-level redundancy structure has r parallel setes, each with k components in series. With independent components, the cdf is

$$F_T(t) = \prod_{i=1}^{r} \left[1 - \prod_{j=1}^{k}(1 - F_{ij})\right]. \tag{15.4}$$

Series-Parallel System with Component-Level Redundancy A $k \times r$ series-parallel system with independent components has k series structures

and r components inin parallel.It has cdf ,

$$F_t(t) = 1 - \prod_{i=1}^{k} \Big(1 - \prod_{j=1}^{r} F_j\Big). \tag{15.5}$$

15.2.5 Bridge System Structure

Depending on if the component at the bridge position works or not, the bridge system can be treated as series-parallel with component level redundancy or system level redundancy, so the cdf can be computed by the sum of two conditional cases.

15.2.6 k-Out-of-s System Structures

A system works if at least k out of s components work but not otherwise. The cdf ,

$$\begin{aligned} F_t(t) &= P(T \leq t) \\ &= P(atleast(s-k+1)components fail) \\ &= \sum_{j=s-k+1}^{s} \Big\{ \sum_{\delta \in A_j} [\prod_{i=1}^{s} F_i^{\delta_i}(1 - F_i^{1-\delta_i})] \Big\}. \end{aligned}$$

15.3 Estimating System Reliability From Component Data

Maximum likelihood estimation for each component and then the mle of the system can be obtained accordingly. Delta method can be used to find the variance-covariance matrix. Normal approximation confidence intervals or bootstrap approximate confidence intervals are proper.

16 Analysis of Repairable System and Other Recurrence Data

16.1 Introduction

16.1.1 Repairable System Reliability Data and Other Recurrence Data

1. Generally, repair times are measured in terms of a system age or time since some well-defined specific event in the system's history.

2. The stochastic model for recurrence data is called a *"point-process" model.*

Repairable system data are collected to estimate or predict quantities like:

1. The distribution of times between repairs, $\tau_j = T_j - T_{j-1}$.
2. The cumulative number of repairs in the interval $(0, t]$ as a function of system age t.
3. The expected time between failures(MTBF).
4. The expected number of repairs in the interval $(0, t]$ as a function of t.
5. The repair rate as a function of t.
6. Average repair cost as a function of t.

16.1.2 A Nonparametric Model for Recurrence Data

For a single system, recurrence data can be expressed as $N(s,\, t)$, in system age interval $(s, t]$, where $N(s, t)$ is the **cumulative number of recurrences in the system**. $N(t)$ is used to represent $N(0, t)$ for simplicity.

1. The model to describe a population of systems is based on the mean cumulative function(MCF) at system age t.
2. The population MCF is defined as $\mu(t) = E[N(t)]$, where the expectation is over the variability of each system and the unit-to-unit variability in the population.
3. $\nu(t) = \frac{\mathrm{d}E[N(t)]}{\mathrm{d}t} = \frac{\mathrm{d}\mu}{\mathrm{d}t}$, is the *recurrence rate* per system for the population given that $\mu(t)$ is differentiable.
4. The method introduced next can also be used to model other quantities accumulating in time than number of repairs, such as the mean cumulative *cost* per system.

16.2 Non-Parametric Estimation of The MCF

16.2.1 Non-Parametric Model Assumptions

If an observed collection of $n \geq 1$ systems is an entire population of interest or a sample from a larger population of systems, then the method described here can be used to estimate the population MCF.

1. There exits a population of cumulative functions(one for each system in the population), from which a sample has been observed.

2. Randomness in the sample is due to the random sampling of cumulative functions from the population.

3. The time that observation of a system is terminated doesn't depend on the system's history.

Biased MCF may be caused by:

1. Units follow a staggered scheme of entry into service and the recurrence rate $\nu(t)$ is increasing in real time due to some possible external events affecting all systems simultaneously. Result: The newer systems that have a more stressful life will be censored earlier and an overly optimistic estimate of the recurrence rate will be obtained. The third assumption is not satisfied.

2. Non-parametric estimator doesn't require that the sampled systems be statistically independent.

16.2.2 Point Estimate of the MCF

A simple naive estimator of the population MCF:
The sample mean of the available $N_i(t)$ for the systems still operating at time t.

- $N_i(t)$ denotes the cumulative number of system recurrence for system i before time t.

- Limitation: this estimator is only appropriate if all systems are still operating at time t.

An appropriate unbiased estimator proposed by Nelson(1988), allows for different lengths of observation among systems.

Algorithm 16.1: Computation of the MCF estimate

1. Order the unique recurrent times, t_{ij} among all n systems, where t_{ij} denotes the jth recurrent time for system i. Let m denote the number of unique times: $t_1 < t_2 < \cdots < t_m$.

2. Compute $d_i(t_k)$, the total number of recurrences for system i at t_k.

3. Let $\delta_i(t_k) = 1$ if system i is still being observed at time t_k, and 0 otherwise.

4. Compute

$$\hat{\mu}(t_j) = \sum_{k=1}^{j} \left[\frac{\sum_{i=1}^{n} \delta_i(t_k) d_i(t_k)}{\sum_{i=1}^{n} \delta_i(t_k)} \right] = \sum_{k=1}^{j} \frac{d_.(t_k)}{\delta_.(t_k)}, \tag{16.1}$$

   where $\delta_.(t_k)$ is the total number of system recurrences at time t_k and $\delta_.(t_k)$ is the size of the risk set at t_k.

One may plot $\hat{\mu}(t)$ as a piecewise linear function for better visual perceptions of shape.

16.2.3 Standard Errors and Non-parametric Confidence Intervals for the MCF

In Nelson(1995a), the true variance of $\mu(t_j)$ for a large population of cumulative functions is:

$$\begin{aligned} Var[\hat{\mu}(t_j)] &= \sum_{k=1}^{j} Var[\bar{d}(t_k)] + 2 \sum_{k=1}^{j-1} \sum_{\nu=k+1}^{j} Cov[\bar{d}(t_k), \bar{d}(t_\nu)] \\ &= \sum_{k=1}^{j} \frac{Var[d(t_k)]}{\delta_.(t_k)} + 2 \sum_{k=1}^{j-1} \sum_{\nu=k+1}^{j} \frac{Cov[d(t_k), d(t_\nu)])}{\delta_.(t_k)}, \end{aligned} \tag{16.2}$$

where $\bar{d}(t_k) = \frac{d_.(t_k)}{\delta_.(t_k)}$ is the average number of recurrences per system at t_k.

Assume that $d_i(t_k)$ is a random sample from the population of $d(t_k)$, and use moment estimation to obtain estimators of the variance and covariance above. More details on Chapter 16.2.3 of Meeker and Escobar(1998).

Pointwise normal-approximation confidence intervals for population MCF can be computed using the general approach and estimated standard error.

When $\mu(t)$ is positive, intervals can be generated based on $[log(\hat{\mu}(t)) - log(\mu(t))]/\hat{se}_{\hat{\mu}(t)} \sim N(0,1)$.

Finite Population Correction: When the number of cumulative functions sampled is more than 5% or 10% of the population, finite population methods should be used for estimating standard errors. In this case,the moment based estimation of variances and covariances should be multiplied by a factor $1 - \frac{\delta.(t_k)}{N}$, where N is the total number of cumulative functions in the population of interest.

16.2.4 Adequacy of Normal-Approximation Confidence Intervals

The adequacy depends on:

1. The number of sample cumulative functions at risk to failure;
2. The shape of the distribution of the cumulative function levels at the point in time where the interval is to be constructed.

Improvements:

1. Use $t_{p;\nu}$ instead of z_p, especially when the number of sample systems at risk is small(say, less than 30).
2. If the cumulative function at a point in time has a normal distribution and if all units are still under observation, then using $t_{p;\nu}$ and $[n/(n-1)]^{1/2} se_{\hat{\mu}}$ provide an exact interval for two or more systems.

16.3 Non-Parametric Comparison of Two Samples of Recurrence Data

Suppose there are two populations/processes with mean cumulative function $\mu_1(t)$ and $\mu_2(t)$, respectively. Let $\Delta_\mu(t) = \mu_1(t) - \mu_2(t)$ represents the difference at time t.

1. A non-parametric estimator of $\Delta_\mu(t)$ is $\widehat{\Delta}_\mu(t) = \hat{\mu}_1(t) - \hat{\mu}_2(t)$.
2. If $\hat{\mu}_1(t)$ and $\hat{\mu}_2(t)$ are independent, an estimate of $Var(\widehat{\Delta}_\mu(t))$ is $\widehat{Var}(\widehat{\Delta}_\mu(t)) = \widehat{Var}(\widehat{\mu}_1(t)) + \widehat{Var}(\widehat{\mu}_2(t))$.
3. Approximate $100(1-\alpha)\%$ confidence interval is based on $Z_{\widehat{\Delta}_\mu} \sim N(0,1)$.

16.4 Parametric Models for Recurrence Data

The most commonly used models for recurrence data are:

1. Poisson process (homogeneous and non-homogeneous).
2. Renewal process.
3. Superimposed versions of above processes.

16.4.1 Poisson Process

An **integer-valued** point process on $[0, \infty)$ is said to be a Poisson process if it satisfies:

1. $N(0) = 0$.
2. The numbers of recurrences in disjoint time intervals are statistically independent.
3. The process recurrence rate $\nu(t)$ is positive and $\mu(a, b) = E[N(a, b)] = \int_a^b \nu(u)\mathrm{d}u < \infty$, when $0 \leq a < b < \infty$.

It follows that $N(a, b)$ has a Poisson distribution:

$$P[N(a, b) = d] = \frac{\mu(a, b)^d}{d!} e^{-\mu(a,b)} \quad d = 0, 1, 2, \ldots \tag{16.3}$$

16.4.2 Homogeneous Poisson Process(HPP)

HPP is a Poisson process with a *constant* recurrence rate, $\nu(t) = \frac{1}{\theta}$.

1. The inter-recurrence times, $\tau_j = T_j - T_{j-1}$, are iid, each with an $exp(\theta)$ distribution, which is followed by $P(\tau_j > t) = p[N(T_{j-1}, T_{j-1} + t) = 0]$.
2. The time $T_k = \tau_1 + \cdots + \tau_k$ to the kth recurrence has a $Gam(\theta, k)$ distribution.

16.4.3 Non-homogeneous Poisson Process(NHPP)

NHPP has a *nonconstant* recurrence rate $\nu(t)$. In this case, the interrecurrence times are neither independent nor identically distributed. The expected number of recurrences per unit time over $(a, b]$ is:

$$\frac{\mu(a, b)}{b - a} = \frac{1}{b - a} \int_a^b \nu(t)\mathrm{d}t. \tag{16.4}$$

An NHPP model is often specified in terms of recurrence rate, $\nu(t) = \nu(t; \boldsymbol{\theta})$.

Two examples:

1. Power model recurrence rate, $v(t; \beta, \eta) = \frac{\beta}{\eta}(\frac{t}{\eta})^{\beta-1}$, $\beta > 0$, $\eta > 0$ with $\mu(t; \beta, \eta) = (\frac{t}{\eta})^{\beta}$.
2. Log-linear model recurrence rate, $\nu(t; \gamma_0, \gamma_1) = \exp(\gamma_0 + \gamma_1 t)$ with $\mu(t; \gamma_0, \gamma_1) = [(\exp)(\gamma_0)][\exp(\gamma_1 t) - 1]/\gamma_1$.

16.4.4 Renewal Process

A sequence of system recurrences at system ages $T_1, T_2, \cdots$ is a renewal process **if the iterrecurrence times τ_j are iid**.

The MCF for a renewal process is known as "renewal function". HPP is a renewal process.

Before using a renewal process, one needs to check for departures from the model:

1. Trend and non-independence of interrecurrence times.
2. If the above assumptions are satisfied, one may use earlier chapters' methods to describe the distribution of interrecurrence times.

16.4.5 Superimposed Renewal Process

The aggregation of renewals from a group of n independent renewal processes operating simultaneously is known as a *superimposed renewal process(SRP)*.

1. Unless the individual renewal processes are HPP, an SRP is NOT a renewal process.
2. When the number of systems n is large and the systems have run long enough to eliminate transients, an SRP behaves as an HPP. (Similar with the central limit theorem).
3. The above result on large number of systems can sometimes be used to justify the use of the exponential distribution to model interrecurrence times.

16.5 Tools for Checking Point-Process Assumptions

16.5.1 Tests for Recurrence Rate Trend

Plots:

1. For a single system, plot of cumulative number of system recurrences vs. time is the simplest plot. Non-linearity in this plot indicates that the interrecurrence times are not identically distributed.

2. A plot of interrecurrence times vs. system age, or a "time series plot" allow the discovery of trends or cycles that would suggest that the interrecurrence times are not identically distributed.

Tests:

1. Power model: "Military Handbook" test for $\beta = 1$(HPP):

$$X^2_{MHB} = -a \sum_{j=1}^{r} \log(t_j/t_{end}), \tag{16.5}$$

   where t_j is the t_{th} recurrence time, and t_{end} is the end of the observation.
   The statistic is approximately $\chi^2_{(2r)}$ under HPP.
   See `http://www.itl.nist.gov/div898/handbook/apr/section2/apr234.htm#TheMilitaryHandbook` for more details.

2. Log linear model: Laplace test for trend in log-linear NHPP model, i.e., $\gamma_1 = 0$,

$$Z_{LP} = \frac{\sum_{j=1}^{r} t_j/t_{end} - r/2}{\sqrt{r/12}}, \tag{16.6}$$

   Z_{LP} has approximate standard normally distribution.

3. Both above tests can be misleading when there is no trend but the underlying process is a renewal process other than HPP. The Lewis-Robinson test for trend,

$$Z_{LR} = Z_{LP} \times \frac{\bar{\tau}}{S_\tau}, \tag{16.7}$$

   where $\bar{\tau}$ is the sample mean and S_{tau} is the standard deviation of the inter-recurrence times. The second term on the right hand side is the *reciprocal of the sample coefficient of variation.*
   In large samples, Z_{LR} follows approximately standard normal distribution under a *renewal process.* Z_{LR} is preferred as a general test of trend in point-process data.

16.5.2 Test for Independent Interrecurrence Times

Consider the serial correlation in the sequence of inter-recurrence times.
Plots: inter-recurrence times vs. lagged inter-recurrence times provides a graphical representation of serial correlation.
Serial correlation coefficient:

$$\rho_k = \frac{Cov(\tau_j, \tau_{j+k})}{Var(\tau_j)}. \tag{16.8}$$

Sample serial correlation:

$$\hat{\rho}_k = \frac{\sum_{j=1}^{r-k}(\tau_j - \bar{\tau})(\tau_{j+k} - \bar{\tau})}{\sqrt{\sum_{j=1}^{r-k}(\tau_j - \bar{\tau})^2 \sum_{j=1}^{r-k}(\tau_{j+k} - \bar{\tau})^2}}, \tag{16.9}$$

where $\bar{\tau} = \sum_{j=1}^{r} \tau_j / r$.

$\sqrt{r-k} \times \hat{\rho}_k \dot{\sim} \text{NOR}(0,1)$ when $\rho_k = 0$ and r is large, and this approximate distribution can be used to asses if ρ_k is different from zero.

16.6 Maximum Likelihood Fitting of Poisson Process

16.6.1 Poisson Process Likelihood

For ONE system in period $(0, t_a]$, we have the data as the number of recurrences, $d_1, \cdots, d_m$ in non-overlapping intervals $(t_0, t_1], \cdots, (t_{m-1}, t_m]$ with $t_0 = 0$ and $t_m = t_a$. The likelihood for the NHPP according (16.3) is,

$$\begin{aligned} L(\boldsymbol{\theta}) &= P[N(t_0, t_1) = d_1, \cdots, N(t_{m-1}, t_m) = d_m] \\ &= \prod_{j=1}^{m} P[N(t_{j-1}, t_j) = d_j] \\ &= \prod_{j=1}^{m} \frac{[\mu(t_{j-1}, t_j; \boldsymbol{\theta})]^{d_j}}{d_j!} \exp^{-\mu(t_{j-1}, t_j; \boldsymbol{\theta})} \\ &= \prod_{j=1}^{m} \frac{[\mu(t_{j-1}, t_j; \boldsymbol{\theta})]^{d_j}}{d_j!} \times \exp^{-\mu(t_0, t_a; \boldsymbol{\theta})}. \end{aligned} \tag{16.10}$$

When the *exact* reported recurrence times at $t_1 \leq \cdots \leq t_r$ ($r = \sum_{j=1}^{m} d_j$), the likelihood in terms of the density approximation is ,

$$L(\boldsymbol{\theta}) = \prod_{j=1}^{m} \nu(t_j; \boldsymbol{\theta}) \times \exp[-\mu(0, t_a; \boldsymbol{\theta})]. \tag{16.11}$$

16.6.2 Superimposed Poisson Process Likelihood

Assumption: all systems have the same $\nu(t)$ (a strong assumption). And systems are independent.

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n}[\prod_{j=1}^{r} \nu(t_{ij};\boldsymbol{\theta}) \times \exp[-\mu(0, t_{a_i};\boldsymbol{\theta})]]. \tag{16.12}$$

As the assumption is strong, it's often inappropriate in practice, and generalizations such as use of explanatory variables to account for system-to-system differences, are possible.

16.6.3 ML Estimation for the Power NHPP and Loglinear NHPP

Plug in the $\nu(t;\boldsymbol{\theta})$ into the likelihood function and solve for the mle and relative likelihood.

16.6.4 Confidence Intervals for Parameters and Functions of Parameters

General ideas in Chapter 7 and 8 can be used.

16.6.5 Prediction of Future Recurrences with a Poisson Process

The expected number of recurrences in an interval $[a, b)$ is $\int_a^b \nu(u;\boldsymbol{\theta})\mathrm{d}u$ and predictions can be made by replacing $\boldsymbol{\theta}$ by the maximum likelihood estimator $\hat{\boldsymbol{\theta}}$.

16.7 Generating Pseudo-Random Realizations from An NHPP Process

Simulation data can be used to check the adequacy of large sample approximation and for implementing bootstrap methods.

16.7.1 General Approach

For a monotone increasing $\mu(t)$, the random variable $\mu(T_{i-1}, T_i)$, $i = 1, \cdots$, are iid, each with exp(1). The idea is to generate random inter-recurrence times and solve for random T values sequentially.

1. Generate U_i, $i = 1, \cdots, r$ from a $Uniform(0, 1)$

$$\begin{aligned}
\mu(T_1) &= -log(U_1) \\
\mu(T_2) - \mu(T_1) &= -log(U_2) \\
&\vdots \\
\mu(T_r) - \mu(T_{r-1}) &= -log(U_r). \qquad (16.13)
\end{aligned}$$

If a realization in an interval $(0, t_a]$ is needed, then r is random and the sequential process is stopped when $T_i > t_a$.
General solution:

$$T_j = \mu^{-1}\big[-\sum_{i=1}^{j} log(U_i)\big]. \qquad (16.14)$$

Or alternatively,

$$T_j = \mu^{-1}\big[\mu(T_{j-1}) - log(U_j)\big], \qquad (16.15)$$

where $T_0 = 0$.

17 Failure-Time Regression Analysis